



PROJECT NARRATIVE AND SPECIFICATIONS

Department of Veterans Affairs
West Palm Beach VAMC
Remove and Replace Public Address System
VA Project No. 548-22-100

100% Construction Documents
December 14, 2022

Prepared by:
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Codes, Standards and Design References

1. VA Sustainable Design Manual, August 2017
2. VA Parking Design Manual, November 2016
3. VA Site Development Design Manual February 2013
4. VA VHA Directive 1061, February 2021
5. VA Fire Protection Design Manual, 2021
6. VA Plumbing Design Manual, May 2021
7. VA HVAC Design Manual, March 2022
8. VA Physical Security and Resiliency Design Manual, September 2021
9. VA Lighting Design Manual, December 2022
10. VA Electrical Design Manual, December 2019
11. VA Interior Design Manual, November 2020
12. VA Infrastructure Standard for Telecommunications Spaces Design Guide, V3.1, July 2021
13. PG 18-13 VA Barrier Free Design Standard, February 2022
14. PG 18-4 VA Standard Details and Specifications
15. NFPA 101, Life Safety Code, 2021
16. NFPA 99, Health Care Facilities Code, 2021
17. NFPA 90A, Installation of Air Conditioning and Ventilating Systems, 2021
18. NFPA 72, National Fire Alarm Signaling Code, 2022
19. NFPA 70, National Electrical Code, 2020
20. NFPA 13, Installation of Sprinkler Systems, 2022
21. ADA Standards for Accessible Design 2010,
22. International Standards for Hematopoietic Cellular Therapy Product Collection, Processing, and Administration, Eighth Edition
23. International Building Code, 2021
24. International Mechanical Code, 2021
25. International Plumbing Code, 2021
26. International Fuel Gas Code, 2021
27. Illuminating Engineering Society (IES) Lighting Handbook, 10th Edition
28. TVHS Energy Requirements and Design Guide, September 2014

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29. Facilities Guidelines Institute: Guidelines for the Design and Construction of Hospitals 2018
30. ASME B31.9, Building Services Piping
31. ASHRAE Standard 170, Ventilation for Health Care Facilities, 2021
32. ASHRAE Handbooks, 2017 Fundamentals, 2019 HVAC Applications, 2020 HVAC Systems and Equipment
33. ASHRAE 90.1 – 2019 Energy Standard for Buildings Except Low-Rise Residential Buildings
34. SMACNA, HVAC Duct Construction Standards, Metal and Flexible
35. SMACNA, HVAC Air Duct Leakage Test Manual
36. ANSI/TIA/EIA 568-2019 Commercial Building Cabling Standards
37. ANSI/TIA/EIA 569-C-2015 Standards for Telecommunications Pathways and Spaces
38. ANSI/TIA-1179-2017, Telecommunications Spaces for Healthcare Facilities
39. BICSI, Telecommunications Distribution Methods Manual, 2020-14 edition
40. Federal Information Processing Standards-FIPS-201-1: Personal Identity Verification (PIV) of Federal Employees and Contractors

General Project Scope

This project will replace the public address system in the VA Medical Center at West Palm Beach, FL, with a Networked Public Address System that is compliant with VA design manuals, design guides and specifications. Project is located within the West Palm Beach VAMC, 7305 N. Military Trail, West Palm Beach, FL 33410.

The new design shall include removing and replacing the existing headend amplifier system located in room BB-263 with a new digital Networked Public Address System. Contractor will be required to remove and replace all components of the existing Public Address System including but not limited to the headend amplifier system & cabinetry in room BB-263, all speakers, components, conductors, conduit, etc. located throughout the facilities. Buildings 1, 3, 15, 16, 18, 19 and 21 are to be included in the new Networked Public Address System.

The architectural scope of work will be limited to ICRA and Life Safety maintenance to install the new Public Address System.

Electrical Design Elements

(Reference *Codes, Standards and Design References* section on Page 1)

The purpose of this project is to upgrade the existing public address and mass notification system (PAMNS) at the VA facility in Palm Beach, Florida. The existing PAMNS has areas of low volume, and areas where pages are not audible. The existing system also is dependent on staff to determine if there are issues with the system.

The Palm Beach VAMC has an existing Simplex paging system. This system will be replaced with a new system utilizing the BiAmp Vocia system as the basis of design. The system will utilize a backbone of ethernet switches connected to the paging system equipment. The electrical scope of this project will be limited to providing emergency sourced electrical outlets to supply headend equipment. Any emergency receptacles added will be sourced from nearby existing critical branch panels.

Telecommunications Design Elements

(Reference *Codes, Standards and Design References* section on Page 1)

The purpose of this project is to replace the existing public address and mass notification system (PAMNS) at the VA Medical Center at West Palm Beach, Florida with a Networked Public Address System that is compliant

The scope of the project includes the remove and replace all components of the existing Public Address System including but not limited to the headend amplifier system and cabinetry in room BB-263, all speakers, components, conductors, conduit etc located throughout the facilities.

The West Palm Beach VAMC has an existing Simplex paging system located in BB-263. This system will be replaced with a new system utilizing the BiAmp Vocia system as the basis of design. New head end equipment will be installed in Equipment Tech. Work Room BC-202. The new system will consist of hardwired paging zones which can be addressed individually or in combination through the control software. This new system will have paging stations located in the call center, energy center, the CLC, the emergency department, and in the police center in the basement. There will also be a system monitoring station located in the basement police center that will show the system status. In addition, the system will be capable of making pages from the campus's VoIP system. It will also be able to announce calls and codes from the campus fire alarm system. The system's status can be monitored from a web app that can be accessed by maintenance personnel.

The system will utilize a backbone of ethernet switches connected to the paging system equipment. The headend equipment will include a headend message server, amplifiers, and any interface equipment. Also attached to the ethernet backbone are end of line devices that track speaker circuits' status. The amplifiers will feed hardwired speaker circuits at 70V using speaker wire. Speaker zones will be laid out by each wing for each floor, and certain operational areas will be provided their own zones (e.g., Emergency department will be placed on its own zone). New head end equipment will be placed adjacent to the existing head end equipment in the basement of building 1, and new paging equipment will be placed in wall mounted cabinets in 3 other locations in building 1 and in the CLC Hospice. The existing head end equipment shall be removed after the new system is installed.

Existing speaker locations will be utilized where possible but may be relocated to obtain an optimal speaker layout. All speakers, speaker cabling and conduit will be replaced. Head end equipment will be installed in existing locations where possible, and new equipment will be installed in locations as needed for system performance and will be independent of VA OIT equipment.

Building 1 is to be included in the new Networked Public Address System.

New 6-strand fiber will be run from the existing basement telecom room to new telecom equipment on the 2nd, 5th, and 8th floors, as well as in the CLC Hospice. Buildings 15, 18, and 21 shall be fed from the head end equipment located in building 1, while building 16 and 19 will be fed from the CLC's head end equipment.

There is an existing parallel fire alarm speaker system which will not be touched as part of this project. The new PAMNS will be interfaced with the fire alarm system to avoid conflicting paging announcements. The system's installation should have no impact on the fire alarm system operation.

PROPOSED PAMN SYSTEM

The new proposed overhead paging system shall have the following:

1. New cabinets to house the new IP based paging system.
2. PoE data network switches.
3. PA message server.
4. Test-to-speech server
5. Digital network two (2) channel amplifier.
6. Digital network four (4) channel amplifier.
7. Digital network multi-channel amplifier chassis type with blade amplifiers as required for each building, floor, and zone.
8. Network audio input devices.
9. Paging stations, desk or wall.
10. Zone modules.
11. Telephone paging adapter.
12. VoIP interface SIP paging adapter
13. Emergency interface server between the PA system and emergency/fire alarm system.
14. Ceiling speakers.
15. Horn speakers.
16. End of line monitoring devices.
17. Cabling.
18. Ambient noise compensation device-adjust output levels automatically in response to changes in ambient noise level in the area.
19. VA OIT Technical Reference Model (TRM) approved software.
20. UPS Back up.
21. NFPA compliant.

The new overhead paging system can do the following:

1. Make an all page to the entire campus.

2. Make a building page.
3. Make a floor page.
4. Make a zone page based on the fire alarm building zone.
5. Make a multi floor page.
6. Make Multi zone page.
7. Make a multi building page.
8. System can be programed to meet the VA needs.
9. System is very flexible.

SYSTEMS OPTIONS

New overhead paging system can also interface/integrate with the following:

1. Integrate with existing NEC DT800 ITZ-12D-3 VOIP.
2. VoIP telephone system through an IP interface that acts as a SIP end point, allowing a direct connection to an existing VoIP call manager. This interface permits live, real time paging using any page code within the new system. The interface supports regular and emergency messaging and will support stacking and store-and forward functions.
3. POTS telephone system through a paging adaptor that allows connections to four analog telephone service lines. This interface permits live, real time paging using any page code within the new system. The interface supports regular and emergency messaging and will support stacking and store-and forward functions.
4. Interface to the existing Siemens emergency/fire alarm system, allowing for the contact closures into the new paging system to mute all audio paths. This immediately allows for onboard audio from the fire alarm, by utilizing all systems that are broadcasting the fire alarm voice evacuation. The full interface allows for fully monitoring and supervised circuits to the DACT on the FACP, allowing strobes to continue but silent horns/speakers when fireman's mic is keyed. This will then supply audio into all the paging circuits into the new system to utilize the paging amplifiers and speakers to hear and understand the evac message.

DATA SYSTEMS

New cabinets for the new overhead paging system shall be provided in Telecommunication Rooms. The cabinets will be grounded.

TELEPHONE SYSTEM INTERFACE-OPTION

The new overhead paging system shall interface with the existing Telephone system. Both POTS (Analog) and VoIP will be available for the interface.

FIRE ALARM SYSTEM INTERFACE-OPTION

The new overhead paging system will be interfaced with the existing fire alarm system in the Police Operations Room BB-204.

ELEVATOR INTERFACE

The new overhead paging system can be interfaced with the existing elevator control panel in the future.

CABLE TRAY/RACEWAYS

1. Cable tray above accessible ceiling shall be used for all low voltage cabling systems except the fire alarm.
2. Cable tray shall be wire mesh type, 12 inch and 24-inch-wide, 4 or 6 inch high.
3. If existing cable is going to be used, remove all existing unused cables before pulling any new cables.
4. Cable tray shall be grounded.
5. Provide 1 ¼ -inch conduit from each data outlet location to each cable tray with plastic bushing.
6. Each IT room shall have 4' X 8' Fire Retarded Plywood per OIT design guide template.

GROUNDING

1. Rack bonding conductor shall be a minimum of # 6 AWG.
2. Cable tray shall be grounded.
3. Ladder tray shall be grounded.
4. Each cabinet and channel rack shall be grounded to the PBB and SBB.

LAN/WAN BACKBONE CABLING

1. A 6/12/24 strand of multi-mode fiber optic cable (OM4) shall be run from each PA cabinet on the serving floor, or from remote building to the main PA head end equipment.
2. Copper cabling shall be provided between PA cabinets that do not exceed the limits per the system manufacturer.

PAMN PROGRAMMING

1. The team shall meet with the PAMN vendor, contractor, subcontractor, VA COR, emergency management team to discuss programming and develop prerecorded messages.

PROJECT PHASING

Most of the PA work will be in the corridors.

We propose that the new PA head end equipment is installed and tested.

We propose that the secondary PA racks for each floor or floors be set and connected to the main PA head equipment.

We propose to remove the existing speakers and cabling from a paging zone, or a floor, install new speakers and wiring and connect speakers to the new PA racks, test zone.

Continue as above for each zone or building until the building is completely moved/switched to the new PA system.

Remove all existing PA system and related material.

Since all work in the corridors and at the nurse's stations will need to be performed while hospital operations are maintained, use of ICRA carts will be made mandatory for all work occurring in the ceiling spaces. Where work for vertical IT drops to control stations occurs, and elsewhere that dust will be produced, temporary soft wall partitions will be used to isolate those areas from the rest of the active hospital area. All work in the corridors and at the nurses' station will be specified to occur at night.

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100% Construction Documents
December 14, 2022

Appendix A

Statement of Probable Cost

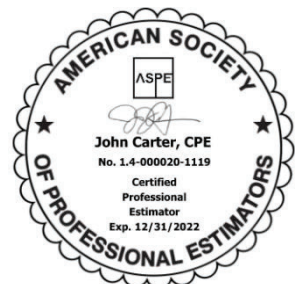


WEST PALM BEACH VAMC
Remove and Replace Public Address System
West Palm Beach, FL

90% Construction Documents – Statement of Probable Cost
October 21, 2022

Prepared by: Evan Jones
Approved by: John J. Carter, CPE

VA Project # 548-22-100
ACG # 22-038



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1. Project Introduction
2. Assumptions / Qualifications
3. Project Data Sheet
4. Cost Estimate:
 - a. Level 1 Construction Cost Summary
 - b. Level 3 Detail
5. Market Survey Report

INTRODUCTION

Project Description

This project will replace the public address system in the VA Medical Center at West Palm Beach, FL, with a Networked Public Address System that is compliant with VA design manuals, design guides and specifications. Project is located within the West Palm Beach VAMC, 7305 N. Military Trail, West Palm Beach, FL 33410.

Basis of Estimate

The estimate has been prepared using the following information:

1. Project Drawings – 90% Construction Documents: Dated: October 21, 2022

<u>GENERAL</u>	<u>LANDSCAPE</u>	<u>FIRE PROTECTION</u>	<u>ELECTRICAL</u>	
G-001	None	None	None	T-114
G-002				T-115
G-003				T-121
G-004				T-122
				T-123
<u>CIVIL</u>	<u>STRUCTURAL</u>	<u>PLUMBING</u>	<u>FIRE ALARM</u>	T-131
None	None	None	None	T-132
				T-133
				T-141
<u>INTERIORS</u>	<u>ARCHITECTURAL</u>	<u>MECHANICAL</u>	<u>TECHNOLOGY</u>	T-142
None	None	None	T-001	T-143
			T-101	T-151
			T-102	T-152
			T-103	T-161
			T-104	T-171
			T-105	T-181
			T-106	T-191
			T-111	T-192
			T-112	T-401
			T-113	T-501
				T-601

2. Project Narrative – 90% Construction Documents: Dated: October 21, 2022

3. Project Specifications – 90% Construction Documents: Dated: October 21, 2022

ASSUMPTIONS / QUALIFICATIONS

Construction Schedule

A construction duration of Twelve (12) Months has been used for the costs for General Conditions. All work is assumed to be during regular business hours.

Method of Procurement

The Estimate is based on traditional Design-Bid-Build procurement competitively bid with a minimum of three (3) qualified SDVOSB General Contractors and a minimum of three (3) qualified subcontractors for major trades required for the project.

Basis for Quantities

Wherever possible, this estimate has been based upon the actual measurement of different items of work. For the remaining items, parametric measurements were used in conjunction with references from other projects of a similar nature.

Basis for Unit Costs

Unit costs are based on current bid prices in the West Palm Beach, FL area or the closest major market of Fort Lauderdale, FL. All subcontractors' overhead and profit are included in each line-item unit cost. This overhead and profit cover each subcontractor's cost for labor burden, materials and equipment sales taxes, field overhead, home office overhead, and profit. The general contractor's overhead and profit is shown separately in the Summary.

Sources for Pricing

This estimate was prepared by qualified cost estimators experienced in estimating construction costs at all design stages. Pricing data to reflect current conditions in the West Palm Beach, FL area. Some quotes may have been solicited from outside sources to substantiate in-house pricing data.

Design Allowances

An allowance for undeveloped design details has been included in the summary of this estimate.

Escalation Allowance

All labor and material prices herein are reflective of current prices. By the VA Manual for Cost Estimates, two separate escalations are included on the summary level to take through the mid-point of construction. Escalation from the date of the estimate to Contract Award assuming the Contract Award is nine (9) months from the date of 100% Sealed Bid Documents using a rate of 8.00% per annum for an escalation of 5.98%.

Escalation from the date of Contract Award to the mid-point of construction is based upon one-half of the estimated construction duration using a rate of 8.00% per annum for an escalation of 4.06%.

Estimated Start date:	July 17, 2023
Midpoint of Construction:	January 15, 2024
Estimated Completion date:	July 16, 2024

Items Excluded From the Estimate, Unless Specifically Noted Otherwise

- Premium time beyond that noted.
- Land acquisition and real estate fees.
- Professional design and consulting fees.
- Owner's field inspection costs.
- General building permit.
- Telephone equipment.
- Move-in costs or maintenance costs after move-in.

Items Affecting the Cost Estimate

Items that may change the estimated construction cost include, but are not limited to:

- Modifications or changes to the scope of work.
- Unforeseen concealed conditions.
- Any specified equipment, material, or product item that cannot be obtained from at least three different sources.
- Any other non-competitive bid situations.
- Bids delayed beyond the projected schedule.

Statement of Probable Cost

This opinion of the probable construction cost is based on the experience, qualifications, and best judgment of an estimator familiar with the construction industry.

PROJECT DATA SHEET

DATE: October 21, 2022

LOCATION: West Palm Beach, FL

PROJECT NO: 548-22-100

TITLE: West Palm Beach VAMC
Remove and Replace Public Address System

Submission No: 90% Construction Documents - Statement of Probable Cost

Area Calculations

719,300 SF Approximate Area of Work

QUANTITATIVE BUILDING DESCRIPTION

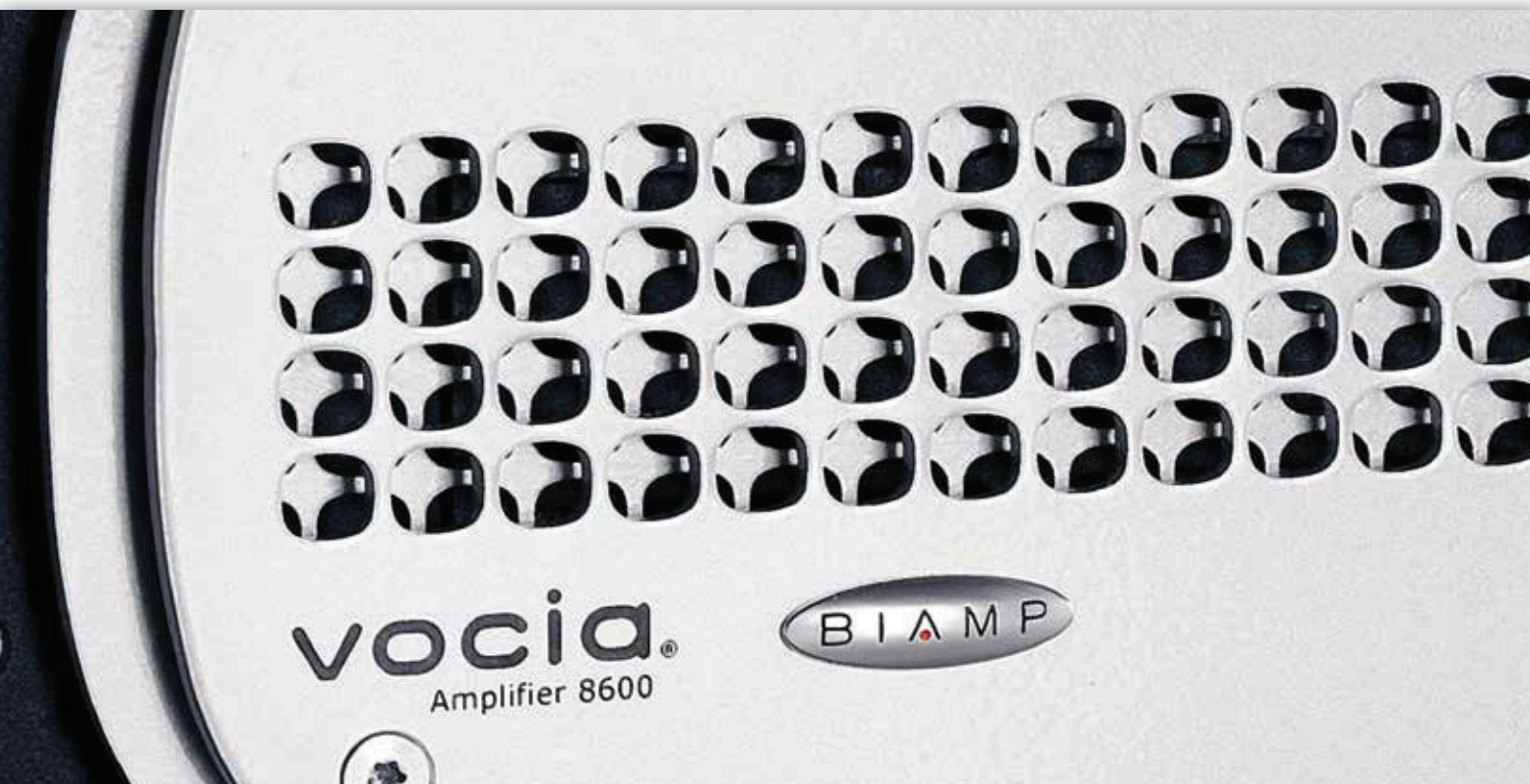
INT. WALL: N/A
EXT. WALL: N/A
WINDOWS: N/A
EXT. DOORS: N/A
INT. DOORS: N/A
NO. OF FLOORS: 9
NO. OF ELEVATORS: N/A

QUALITATIVE BUILDING DESCRIPTION (TYPE & MATERIAL)

FOUNDATION: Existing
SUBSTRUCTURE: Existing
SUPERSTRUCTURE: Existing
EXTERIOR CLOSURE: Existing
ROOFING: Existing
INTERIOR WALLS: Existing

Appendix D

PA Cutsheets



VOCIA AMPLIFIER 8600

EFFECTIVE JANUARY 2015

PRODUCT OFFERINGS

VOCIA®

Headquarters

BIAMP SYSTEMS
9300 S.W. Gemini Drive
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Europe Central
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South America
southamerica@biamp.com

Middle East & Africa
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CONTACT SUPPORT
support@biamp.com

To order product catalogs and other supporting literature, visit our Literature Ordering Link Online (LOLO). Go to www.biamp.com, click on My Account and look for the Order Literature (LOLO) link in the right hand column.

Or, to inquire about orders contact: orders@biamp.com

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AUDIO INPUTS [5]

Vocia input devices are available for both desk stations and wall stations, allowing flexibility in user location.

AUDIO OUTPUTS [10]

Vocia output devices provide both flexible and scalable functionality.

CONTROLLERS [15]

Vocia controllers complement varying system sizes and needs, from output level control to emergency/fire alarm interfaces.

SERVERS [16]

Vocia servers provide user-controlled audio, including global paging and text-to-speech functionality.

SOFTWARE [18]

From an individual world to a multi-world universe, Vocia software is intuitive and easy to use.



MEET THE NEXT GENERATION

Vocia is more than a basic public address system. It's the result of Biamp understanding evolving communication trends and the demands put on today's systems. By using proven IT technologies and applying them to mass communication needs, Vocia delivers a superior alternative to the traditional, analog public address system. No matter how challenging your needs or circumstances, Vocia delivers a system to meet your requirements, today and in the future.

EASILY SCALABLE, FROM SIMPLE TO SPECTACULAR

Vocia fits a wide range of installation sizes and configurations because every component contains memory and processing resources that communicate intelligently across your network. Its modular hardware and easy-to-use software help you imagine and implement the right approach for a simple overhead paging application or a complex system with many zones—and everything in between.

With Vocia, it's also a snap to expand the reach and feature set in your installation at any time. Vocia can do more and demand less of you and your end users.

GOODBYE, COMPLEXITY. HELLO, CONFIDENCE.

Say farewell to the traditional centralized system approach. And the limitations that come with it. Vocia distributes message routing and processing across the network, so it eliminates the chance that a single failure can bring down your system. You can monitor, log and control components remotely, and plug and play units, expanding your system as needed. In short: your maintenance duties scale down as your system capabilities scale up.

With Vocia, you can be confident that your networked public address and voice evacuation system will do its job here and now, as well as into the future.

MORE RELIANCE ON YOUR SYSTEM

Vocia was designed to do more than deliver clear information, it was designed with an understanding of the exacting system requirements around the globe. As the first U.S. manufacturer to earn EN 54-16 certification, you gain peace of mind that your system is backed by quality, commitment to excellence, and reliability.

Vocia builds on network standards you already know, including CobraNet® for single-site installations and TCP/IP for multi-site installations. You simply extend your existing practices for network redundancy, security and monitoring to your paging and voice evacuation system.

BUILT ON DEPENDABLE BIAMP QUALITY AND RELIABILITY

Since 1976, organizations around the world have depended on Biamp to deliver reliable and useful innovation. No matter where you are in the world, you'll find Biamp technology working within audio systems in health care, higher education, transportation, government, event facilities and other demanding environments. With Vocia, you'll get the power to move forward with a distributed approach to public address and voice evacuation.

With Vocia at the heart of your system, you can always count on sound that moves people.

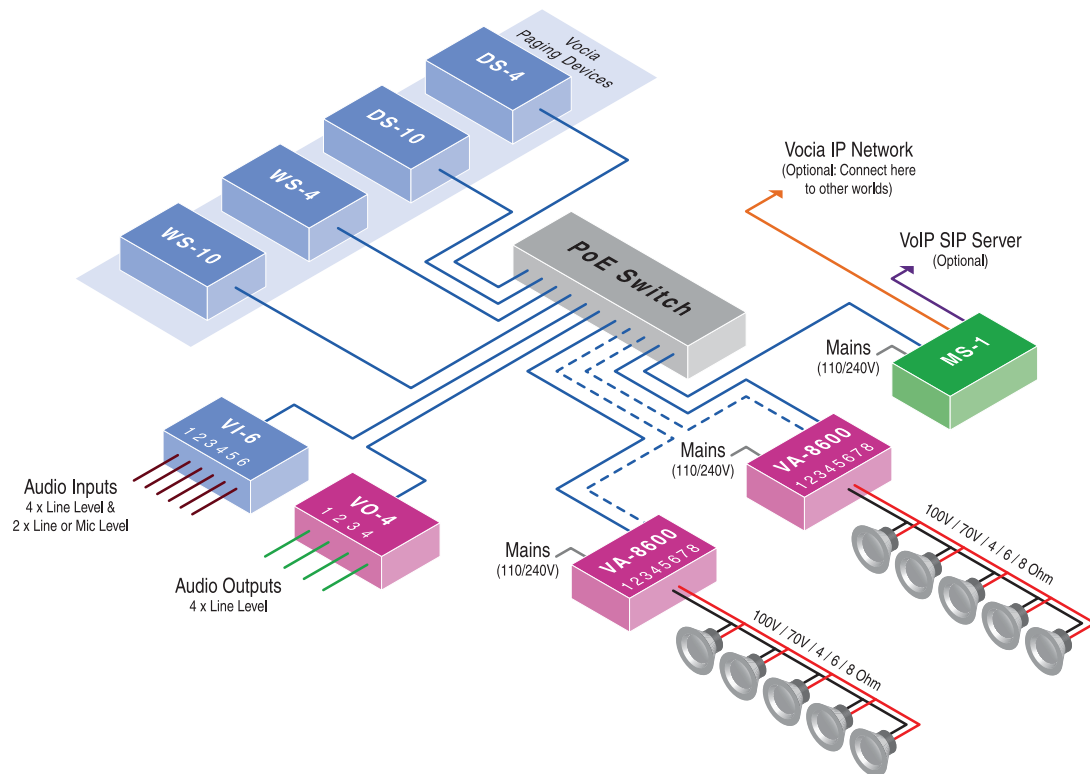
For more information, product details and case studies, visit www.biamp.com/vocia

SYSTEM OVERVIEW

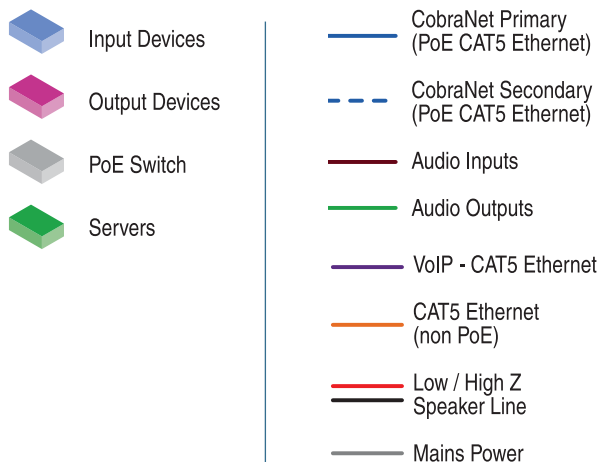
VOCIA® SYSTEM OVERVIEW

As one of Biamp's Networked Media Systems, Vokia, provides facilities of all sizes with efficient, flexible public address and voice evacuation systems with the security and dependability you need.

Vokia offers a high degree of scalability, from simple overhead paging applications to advanced, networked public address systems spanning multiple zones and structures. With decentralized processing and page routing, Vokia's smart units use network technology to distribute intelligence—eliminating the potential for a single point of system failure. And with Biamp's commitment to meeting the various system needs around the world, we built Vokia from the ground up to meet the exacting requirements for Voice Alarm Control and Indicating Equipment (VACIE).



This diagram illustrates a standard, single-world Vokia system installation.



AUDIO INPUTS



The DS-10 is a desktop networked paging station featuring embedded DSP and on-board memory, supporting both standard and advanced public address functionality. All device-specific configuration information is stored locally, which means the DS-10 does not rely on a centralized controller for processing and page routing.

- Push-to-talk button with status indication
- Up to 999 user-configurable page codes
- Supports up to 250 software configurable non-emergency priority paging levels
- Local digital signal processing, including gain, filters and compressor/limiter
- Auxiliary port provides connection to VAM-1 or VPSI-1 device
- RS-232 connection for transmitting Vocia Text Protocol
- Built-in auto repeat
- Local storage of default and/or custom preambles
- Built-in store and forward functionality
- CobraNet audio/control with dynamic use of available bundles, plus power over a single cable
- Backlit liquid crystal display (LCD)
- Optional PIN to restrict unauthorized use
- High-quality gooseneck cardioid dynamic microphone with dual transducer (monitored)
- Rotary ID switches for unit identification
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

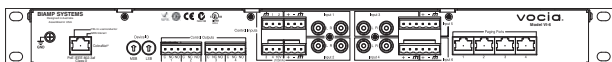
DS-10
VOCIA DESK STATION



The DS-4 is a desktop networked paging station featuring embedded DSP and on-board memory, supporting both standard and advanced public address functionality. All device-specific configuration information is stored locally, which means the DS-4 does not rely on a centralized controller for processing and page routing.

- Push-to-talk button with status indication
- 4 user-configurable page codes
- Supports up to 250 software configurable non-emergency priority paging levels
- Local digital signal processing, including gain, filters and compressor/limiter
- Auxiliary port provides connection to VAM-1 or VPSI-1 device
- RS-232 connection for transmitting Vocia Text Protocol
- Built-in auto repeat
- Local storage of default and/or custom preambles
- Built-in store and forward functionality
- CobraNet audio/control with dynamic use of available bundles, plus power over a single cable
- Backlit liquid crystal display (LCD)
- Optional PIN to restrict unauthorized use
- High-quality gooseneck cardioid dynamic microphone with dual transducer (monitored)
- Rotary ID switches for unit identification
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

DS-4
VOCIA DESK STATION



The VI-6 is a networked audio input expansion device allowing the user to add six channels of background music or user-defined audio to a Vocia system. As part of the Vocia system, the VI-6 meets paging requirements for facilities of all sizes.

- 4 sets of dual RCA connectors, plus terminal block connectors for line-level inputs
- Supports up to 4 paging inputs from VAM-1 and VPSI-1 devices
- 4 control inputs and 4 control outputs
- 2 microphone/line inputs with phantom power
- Software-configurable local audio signal processing, including gain, filters and compressor/limiter
- Rotary switches for unit identification
- CobraNet audio/control with dynamic use of available bundles, plus power over a single cable
- Status LEDs to indicate signal and clip
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VI-6
VOCIA INPUT

AUDIO INPUTS

WS-4
VOCIA WALL STATION



The WS-4 is a wall-mounted networked paging station featuring embedded DSP and on-board memory to support both standard and advanced public address functionality. All device-specific configuration information is stored locally, which means the WS-4 does not rely on a centralized controller for processing and page routing.

- Push-to-talk button with status indication
- 4 user-configurable page codes
- Supports up to 250 software configurable non-emergency priority paging levels
- Local digital signal processing, including gain, filters and compressor/limiter
- Auxiliary port provides connection to VAM-1 or VPSI-1 device
- RS-232 connection for transmitting Vocia Text Protocol
- Built-in auto repeat
- Local storage of default and/or custom preambles
- Built-in store and forward functionality
- CobraNet audio/control with dynamic use of available bundles, plus power over a single cable
- Backlit liquid crystal display (LCD)
- Optional PIN to restrict unauthorized use
- High-quality, noise-cancelling handheld microphone with dual transducer (monitored)
- Rotary ID switches for unit identification
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

WS-10
VOCIA WALL STATION



The WS-10 is a wall-mounted networked paging station featuring embedded DSP and on-board memory to support both standard and advanced public address functionality. All device-specific configuration information is stored locally, which means the WS-10 does not rely on a centralized controller for processing and page routing.

- Push-to-talk button with status indication
- Up to 999 user-configurable page codes
- Supports up to 250 software configurable non-emergency priority paging levels
- Local digital signal processing, including gain, filters and compressor/limiter
- Auxiliary port provides connection to VAM-1 or VPSI-1 device
- RS-232 connection for transmitting Vocia Text Protocol
- Built-in auto repeat
- Local storage of default and/or custom preambles
- Built-in store and forward functionality
- CobraNet audio/control with dynamic use of available bundles, plus power over a single cable
- Backlit liquid crystal display (LCD)
- Optional PIN to restrict unauthorized use
- High-quality, noise-cancelling handheld microphone with dual transducer (monitored)
- Rotary ID switches for unit identification
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty



The EWS-4 is the emergency, wall-mounted networked paging station used in an EN 54-16 certified life safety system. It features all the same configurable page codes, memory and features as the WS-4 paging station.

- Includes all functionality of the WS-4 paging station
- Push-to-talk button with status indication
- 4 user-configurable Emergency Page Codes
- Supports up to 250 software configurable Emergency paging priority levels
- Built-in store and forward functionality
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

EWS-4
EMERGENCY WALL STATION
EN 54-16 CERTIFIED



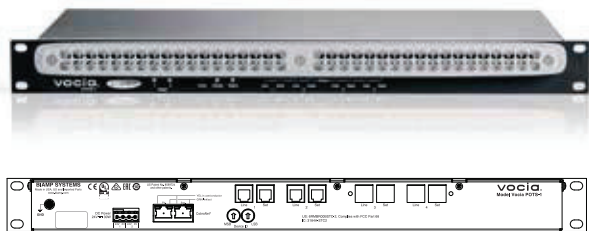
The EWS-10 is the emergency, wall-mounted networked paging station certified for use in an EN 54-16 compliant life safety system. It features all the same configurable page codes, memory and features as the WS-10 paging station.

- Includes all functionality of the WS-10 paging station
- Push-to-talk button with status indication
- Up to 999 user-configurable Emergency Page Codes
- Supports up to 250 software configurable Emergency paging priority levels
- Built-in store and forward functionality
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

EWS-10
EMERGENCY WALL STATION
EN 54-16 CERTIFIED

AUDIO INPUTS

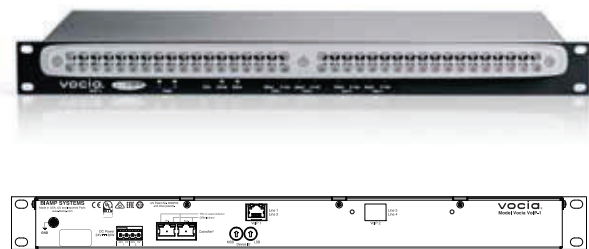
POTS-1 VOCIA TELEPHONE PAGING ADAPTER



The Vocia Telephone Paging adapter allows a direct connection of two or four analog telephone service (POTS) lines or POTS-based PBX telephone lines. The interface permits live, real-time paging using any page code within the Vocia system. POTS-1 supports Vocia regular and emergency messaging, and will also support page stacking and store-and-forward functionality.

- Two or four POTS lines (POTS-1-2 or POTS-1-4)
- Device monitoring
- Power via dual 24V DC inputs (redundant inputs)
- Software-configurable signal processing
- Rack mountable (1RU)
- Dual CobraNet® ports for redundant Ethernet connections
- Rotary switches for device identification
- IP30 compliant
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

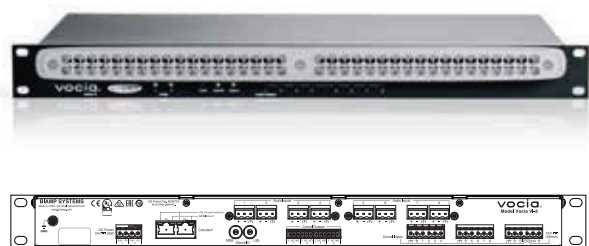
VOIP-1 VOCIA VOICE OVER IP INTERFACE



The Vocia Voice over IP Interface (VoIP-1) acts as a SIP end point allowing a direct connection to an existing VoIP call manager. The interface permits live, real-time paging using any page code within a Vocia system. The VoIP-1 supports Vocia regular and emergency messaging, and will also support page stacking and store-and-forward functionality.

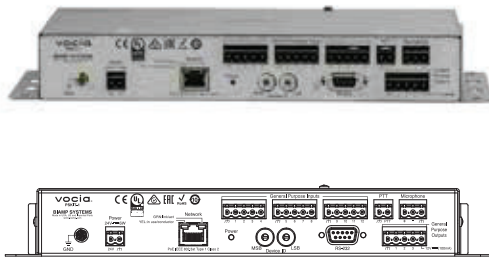
- Two or four VoIP lines (VOIP-1-2 or VOIP-1-4)
- Multiple codec support
- Device monitoring
- Power via dual 24V DC inputs (redundant inputs)
- Software-configurable signal processing
- Rack mountable (1RU)
- Dual CobraNet ports for redundant Ethernet connections
- Rotary switches for device identification
- IP30 compliant
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VI-8 VOCIA 8-CHANNEL LIVE AUDIO INPUT DEVICE



The Vocia Input 8 (VI-8) is designed to facilitate live audio paging from user sources to emergency and non-emergency zones in a Vocia system. The VI-8 will allow direct paging access from eight sources via analog mic/line level inputs or CobraNet digital audio connections. Nine logic inputs can be assigned to various Vocia control events including paging control. A logic output and four relay outputs can be assigned to control events. The logic output may be used to report status from one or several daisy-chained VI-8s to an external system. The logic inputs and output are monitored.

- 8 analog mic/line level or CobraNet inputs
- 1 logic and 4 relay control outputs
- 8 logic inputs
- Device status monitoring via control logic input and output
- Power via dual 24V DC inputs (redundant inputs)
- Software-configurable signal processing including volume control, filters, compressor/limiting
- Rack mountable (1RU)
- Dual CobraNet ports for redundant Ethernet connections
- Device and logic I/O status monitoring
- Rotary switches for device identification
- IP30 compliant
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty



The PSKIT-1 is a standalone Paging Station Kit designed to allow for direct connection to third-party equipment such as fireman's microphone stations or custom designed fire panels. The PSKIT-1 features embedded DSP and on-board memory to support standard and advanced public address and mass notification functionalities. The PSKIT-1 can store 999 user-configurable Page Codes. Additionally, all device-specific configuration information is stored locally, which means the PSKIT-1 does not rely on a centralized controller for processing and page routing.

- Up to 999 user-configurable Page Codes
- Supports up to 250 software configurable priority paging levels
- Can be configured to perform as a EWS or WS microphone
- Local storage of default and/or custom preambles
- Built-in store and forward functionality
- Microphone and Push-to-Talk (PTT) input
- Dual/Redundant Powering (PoE, 24V DC)
- 12 General Purpose Inputs
- 3 General Purpose Outputs
- Rotary ID switches for unit identification
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

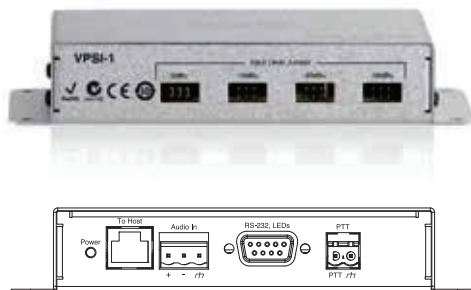
PSKIT-1
VOCIA PAGING STATION KIT



The VAM-1 device is an independent microphone assembly that functions as a slave device to the Vocia Wall and Desk Station (WS-4/10 and DS-4/10) series microphones and to the Vocia Input 6 (VI-6) for paging via the Paging Ports. Up to four VAM-1 slave microphones can be connected per VI-6. The VAM-1 incorporates a PTT switch and has LED indication of the Wait, Talk Now and Unavailable paging states. The microphone latch is magnetic for easy docking to the cradle. Power is provided by the host device.

- Slave device to non-emergency Vocia paging stations and VI-6 devices
- Visual feedback of Paging and Zone status
- Suitable for mounting on a wall or desk
- Audio and Control over a single Ethernet cable
- IP30 Compliant
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VAM-1
VOCIA AUXILIARY MICROPHONE



The VPSI-1 device is designed to allow third party microphones and LED indicators to interface as slave devices to the Vocia standard and emergency Wall Station and standard Desk Station (WS-4/10, EWS-4/10 and DS-4/10) series microphones and to the Vocia Input 6 (VI-6) device for paging via the Paging Ports. The VPSI-1 can also facilitate third party microphones and control system interfacing with Vocia Wall and Desk Stations. The host Paging Station must be configured for 'Remote control' mode within the software.

- Slave interface to Vocia Paging Stations and VI-6 device
- Supports RS-232 connection for transmitting Vocia Text Protocol or LED paging indication mode
- Visual feedback of Paging and Zone status
- Suitable for surface mounting
- Audio, Power and Control over a single Ethernet cable
- IP30 Compliant
- Power is provided by the host device
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VPSI-1
VOCIA PAGING STATION INTERFACE

AUDIO OUTPUTS

VO-4 VOCIA OUTPUT



The VO-4 is a networked audio output expansion device allowing the user to add four line-level output channels to a Vocia system. The VO-4 accepts four channels of digital audio input via CobraNet and provides four line-level analog audio outputs. The VO-4 features embedded DSP and on-board memory to process and store all device-specific configuration information locally and includes comprehensive fixed-chain digital signal processing.

- Converts digital audio to analog audio
- 4 removable terminal block connectors for line-level outputs
- 4 control inputs and 4 control outputs
- Software-configurable local audio signal processing, including gain, filters and compressor/limiter
- Rotary switches for unit identification
- Power over Ethernet (PoE)
- CobraNet audio/control with dynamic use of available bundles, plus power over single Ethernet cable
- Status LEDs to indicate signal and clip
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VO-4e VOCIA OUTPUT



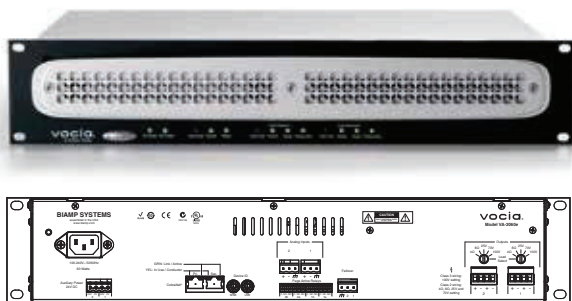
The VO-4e is an enhanced networked audio output expansion device allowing the user to add four line-level and CobraNet output channels to a Vocia system. The VO-4e can be configured for channel-to-channel or device-to-device failover and uses comprehensive fixed-chain digital signal processing within the device, including volume control, ducking, equalization, compressor/limiter, speaker crossover, delay, and output gain. Emergency messages for life safety systems are stored in non-volatile memory. Two RJ45 connectors on the rear panel of the device provide redundant connectivity to process control data, audio and power over a single Ethernet cable. In addition to this the VO-4e also has dual inputs for accepting power from an auxiliary supply. The per-channel paging relay provides a contact closure when paging is active on an associated channel.

- CobraNet audio output
- Channel-to-channel and device-to-device failover
- Local non-volatile storage of Emergency Messages
- Dual PoE capable with alternate powering from auxiliary 24V DC supply (dual inputs)
- LED status and fault indication
 - Chassis Fault, Activity and Status
 - PoE Power
 - Aux Power
 - Output Channel Amp Fault, Activity and Signal Present
- Four Page Active Relays, Control Inputs and Control Outputs
- Speaker line monitoring and ambient noise compensation using ELD-1 and ANC-1
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization, and output level
- CobraNet audio/control with dynamic use of available bundles over single Ethernet cable
- Dual CobraNet ports for redundancy
- Rotary switches for device identification
- IP30 Compliant
- CE marked, UL listed and RoHS compliant
- Rack mountable (IRU)
- Covered by Biamp Systems' five-year warranty



The VA-2060 is a digital networked two-channel amplifier. It is CobraNet enabled and capable of delivering continuous audio power at 60 Watts RMS per channel. The VA-2060 also has comprehensive failover capability with device-to-device and channel-to-channel failover.

- Two channels of 60W RMS per channel
- Transformer coupled outputs, low impedance (4 or 8 Ω) or 25V, 70V or 100V (hardware selectable)
- Comprehensive failover
 - Device-to-device
 - All-to-1 channel
 - 1:1 channel
- Supports Page Active Relay (PAR), ELD-1 and ANC-1
- Local non-volatile storage of emergency messages
- LED status and signal indication
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization and output sensitivity
- CobraNet audio/control with dynamic use of available bundles
- Dual Ethernet ports for redundancy
- Rotary switches for device identification
- Rack mountable (2RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

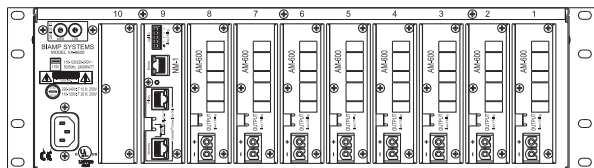


The VA-2060e is a digital networked two-channel amplifier. It is CobraNet enabled and capable of delivering continuous audio power at 60 Watts RMS per channel. It has local analog inputs and dual power supply. The VA-2060e also has comprehensive failover capability with device-to-device and channel-to-channel failover.

- Two channels of 60W RMS per channel
- Transformer coupled outputs, low impedance (4 or 8 Ω) or 25V, 70V or 100V (hardware selectable)
- Comprehensive failover
 - Device-to-device
 - All-to-1 channel
 - 1:1 channel
- Supports Page Active Relay (PAR), ELD-1 and ANC-1
- Local non-volatile storage of emergency messages
- LED status and signal indication
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization and output sensitivity
- CobraNet audio/control with dynamic use of available bundles
- Local background source support with single mic/line analog audio input for each channel (gain-adjustable, switchable phantom power)
- Dual power supply
 - AC Mains
 - Single or Dual 24V DC inputs
- Dual Ethernet ports for redundancy
- Rotary switches for device identification
- Rack mountable (2RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

AUDIO OUTPUTS

VA-8600
VOCIA AMPLIFIER



no more than 2400 watts.

The VA-8600 is a networked multi-channel amplifier. It is CobraNet enabled and features eight channels of modular amplification and DSP with optional channel-to-channel or device-to-device failover.

- Modular based design
- Amplification modules with software configurable power levels/load options
 - 8 amplification modules per frame with 100 to 600 Watts per module (maximum of 2400W peak per chassis)
 - 70V or 100V with direct drive capability, or low-impedance (4 Ω or 8 Ω) operation
- Maximum of 2400 Watts of power in a device
- Failover capability between channels and amplifiers
- Local non-volatile storage of emergency messages
- LED indication:
 - Amplifier failure
 - Clip present
 - Fan stuck-rotor
 - Heat sink temperature fault
 - Signal peak
 - Signal present
- Software monitoring features:
 - Amplifier failure
 - Excessive clipping
 - Fan stuck-rotor
 - Heat sink temperature fault
 - Peak present
 - Short circuit on output
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization and output sensitivity
- CobraNet audio and control data over a single Ethernet cable
- Dual Ethernet ports for redundancy
- Heavy duty removable terminal block connectors for speaker line connections
- Rotary switches for unit identification
- Rack mountable (3RU)
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VA-8600c
VOCIA AMPLIFIER
EN 54-16 CERTIFIED



The VA-8600c is a networked multi-channel amplifier. It is CobraNet enabled and features eight channels of modular amplification and DSP with optional channel-to-channel or device-to-device failover.

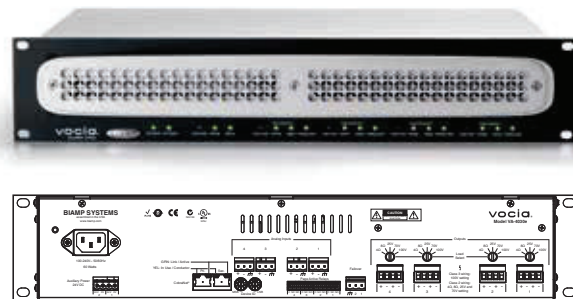
VA8150cv amp



The VA-4030 is a digital networked four-channel amplifier. It is CobraNet enabled and capable of delivering continuous audio power at 30 Watts RMS per channel. The VA-4030 also has comprehensive failover capability with device-to-device and channel-to-channel failover.

- Four channels of 30W RMS per channel
- Transformer coupled outputs, Low impedance (4 or 8 Ω) or 25V, 70V or 100V (hardware selectable)
- Comprehensive failover
 - Device-to-device
 - All-to-1 channel
 - 1:1 channel
- Supports Page Active Relay (PAR), ELD-1 and ANC-1
- Local non-volatile storage of emergency messages
- LED status and signal indication
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization and output sensitivity
- CobraNet audio/control with dynamic use of available bundles
- Dual Ethernet ports for redundancy
- Rotary switches for device identification
- Rack mountable (2RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

VA-4030
VOCIA AMPLIFIER
EN 54-16 CERTIFIED



The VA-4030e is a digital networked four-channel amplifier. It is CobraNet enabled and capable of delivering continuous audio power at 30 Watts RMS per channel. It has local analog inputs and dual power supply. The VA-4030e also has comprehensive failover capability with device-to-device and channel-to-channel failover.

- Four channels of 30W RMS per channel
- Transformer coupled outputs, low impedance (4 or 8 Ω) or 25V, 70V or 100V (hardware selectable)
- Comprehensive failover
 - Device-to-device
 - All-to-1 channel
 - 1:1 channel
- Supports Page Active Relay (PAR), ELD-1 and ANC-1
- Local non-volatile storage of emergency messages
- LED status and signal indication
- Software-configurable signal processing including volume control, filters, compressor/limiting, delay, speaker equalization and output sensitivity
- CobraNet audio/control with dynamic use of available bundles
- Local background source support with single mic/line analog audio input for each channel (gain adjustable, switchable phantom power)
- Dual power supply
 - AC Mains
 - Single or Dual 24V DC inputs
- Dual Ethernet ports for redundancy
- Rotary switches for device identification
- Rack mountable (2RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked, UL listed & RoHS compliant
- Covered by Biamp Systems' five-year warranty

VA-4030e
VOCIA AMPLIFIER
EN 54-16 CERTIFIED

AUDIO OUTPUTS

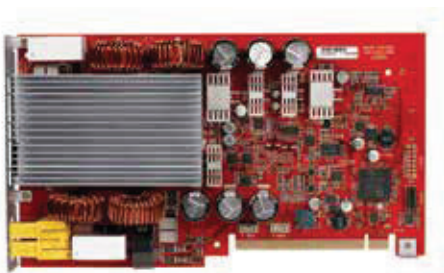
AM-600
VOCIA AMPLIFIER
MODULE



AM-600: Basic Amplifier Module for VA-8600.

- 100 to 600 Watts per module
- Low impedance (4, 6, 8 Ω), 70-Volt or 100-Volt line direct drive
- Power levels/load options are software configurable
- Front-panel LED indication for module fault, activity and signal presence
- UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

AM-600c
VOCIA AMPLIFIER MODULE
EN 54-16 CERTIFIED



AM-600c: Amplifier module with standards-compliant ground fault detection for VA-8600.

- 100 to 600 Watts per module
- Low impedance (4, 6, 8 Ω), 70-Volt or 100-Volt line direct drive
- Power levels/load options are software configurable
- Front-panel LED indication for module fault, activity and signal presence
- Reports fault upon detection of short to ground on speaker line, where this short would compromise capability of amplifier to deliver emergency messages
- EN 54-16 certified, EN 60849 & AS 60849 verified, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

IM-16
VOCIA INTERFACE MODULE
EN 54-16 CERTIFIED



IM-16: The Interface Module 16 physically fits into the LSI-16 option slot. The IM-16 offers an additional 16 control inputs that can be configured in the Vocio software as alarm, fault or reset inputs.

- 16 control inputs
- Each input can be programmed as TTL, high range or monitored high range
- Monitoring for short-to-ground, open circuit and over voltage
- Suitable for the LSI-16 option slot
- Power, processing and indication by the LSI-16
- Cable connections via pluggable screw terminal blocks
- EN 54-16 certified, EN 60849 & AS 60849 verified, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

PARM-1
VOCIA PAGE ACTIVE
RELAY MODULE



PARM-1: Page Active Relay Module for VA-8600 (optional).

- Allows control of external devices during paging (e.g. attenuator over-ride relays)
- Suitable for the VA-8600 Option Slot
- Powered from and controlled by the VA-8600
- 8 relay circuits
- Connections via 3.5mm pluggable screw terminal blocks
- UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

VFOM-1
VOCIA VA-8600 FAILOVER
MODULE EN 54-16 CERTIFIED



VFOM-1: The failover module is for use in the VA-8600/VA-8600c amplifiers (optional). It offers selectable 7:1 or dual 3:1 channel failover.

- 7:1 channel failover
OR
Two sets of 3:1 channel failover
- EN 54-16 certified, EN 60849 & AS 60849 verified, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

CONTROLLERS



The ANC-1 allows automatic output level adjustments in response to changes in ambient noise levels. The ANC-1 is a networked device for use with the Vocol VA-8600. The Vocol software interface permits comprehensive adjustment of ANC parameters and utilizes IEEE compliant Power over Ethernet (PoE) technology.

- Automatic, adaptive volume adjustment based on ambient noise sensing and processing
- 48V phantom power
- Wall mountable
- CobraNet audio/control with dynamic use of available bundles, plus power on a single Ethernet cable
- Status LEDs
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

ANC-1
VOCIA AMBIENT NOISE
COMPENSATION DEVICE



The ELD-1 is a networked safety device for use with the VA-8600 amplifier and is an integral part of standards-compliant voice evacuation and paging notification systems.

- Line monitoring
- Looks for inaudible signal from Vocol amp module
- Reports shorts or opens in the speaker line to the network
- Surface mountable
- Power and data over a single Ethernet cable
- Status LED
- Removable terminal block connector
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

ELD-1
VOCIA END OF LINE DEVICE



The GPIO-1 provides 16 general purpose inputs and outputs to control various aspects of a Vocol system. The GPIO-1 is a monitored device and can be used with an LSI-16e in life safety applications where more logic inputs or outputs are required. The GPIO-1 has dual powering from PoE Ethernet ports and alternate powering from dual 24V DC inputs. In the event of power loss, changeover between power sources will provide uninterrupted operation.

- 16 general purpose logic inputs and outputs
- Device monitoring
- Able to be used to directly interface with fire alarm and emergency equipment
- PoE capable with alternate powering from auxiliary 24V DC supply (dual inputs)
- Software-configurable
- Control over a single Ethernet cable
- Dual Ethernet ports for redundancy
- Rotary switches for device identification
- IP30 Compliant
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

GPIO-1
VOCIA GENERAL PURPOSE
INPUT/OUTPUT

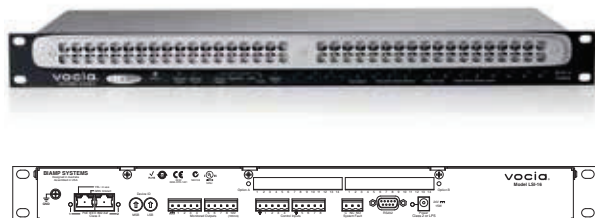


The WR-1 is a networked wall remote designed to control background audio in user-defined music zones. Utilizing Power over Ethernet (PoE) technology, it allows users to select background music sources, inhibit pages and mute sources—all from an attractive, low-profile, wall-mounted panel.

- Wall mountable (US 2-gang)
- Power and data over a single Ethernet cable
- Backlit liquid crystal display (LCD)
- Rotary switches for unit identification
- Software-configurable settings, including volume, source selection, paging inhibit and mute source control
- CE marked and RoHS compliant
- Covered by Biamp Systems' five-year warranty

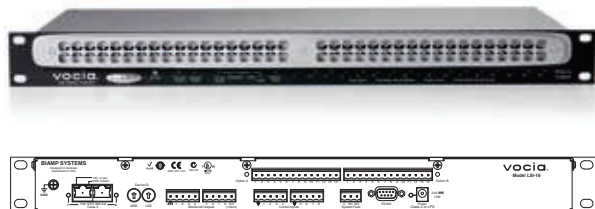
WR-1
VOCIA WALL REMOTE

SERVERS

LSI-16
VOCIA LIFE SAFETY INTERFACE
EN 54-16 CERTIFIED

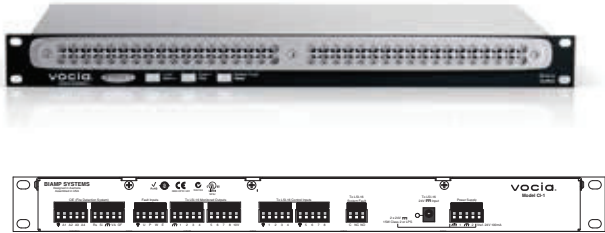
The LSI-16 serves as an emergency interface between a Vocola system and emergency/fire alarm systems. While typically powered from a certified 24V DC source, it can also utilize Power over Ethernet (PoE).

- Parallel I/O ports for direct interface with fire alarm and emergency equipment
- 8 monitored I/O and 8 control inputs
- Redundant network connection and power supply options
- Power and data over a single Ethernet cable
- Local storage of configuration data
- Rotary switches for unit identification
- Accepts the Interface Module 16 (IM-16) for 16 additional general purpose inputs
- Status LEDs
- Rack mountable (1RU)
- Up to 4 discrete emergency inputs
- Up to 500 virtual inputs via RS232 port or Ethernet
- EN 54-16 certified, EN 60849 & AS 60849 verified, UL listed and RoHS compliant
- Covered by Biamp Systems' five-year warranty

LSI-16e
VOCIA LIFE SAFETY INTERFACE
EN 54-16 CERTIFIED

The LSI-16e is an enhancement to the LSI-16 adding 16 additional control inputs. The LSI-16e serves as an enhanced emergency interface between a Vocola system and emergency/fire alarm systems. While typically powered from a certified 24V DC source, it can also utilize Power over Ethernet (PoE).

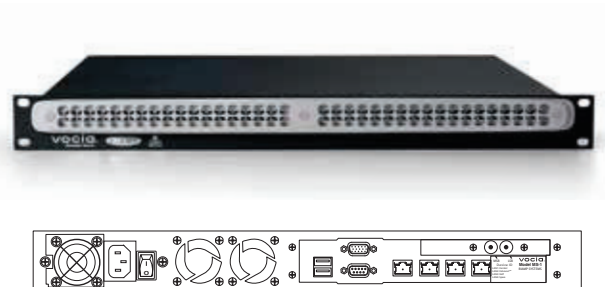
- Parallel I/O ports for direct interface with fire alarm and emergency equipment
- 8 monitored I/O and 8 control inputs
- Redundant network connection and power supply options
- Power and data over a single Ethernet cable
- Local storage of configuration data
- Rotary switches for unit identification
- Up to 4 discrete emergency inputs
- 16 additional general purpose inputs can be programmed to play emergency message, enable zone reset or zone silence; maximum 10 inputs can be assigned per emergency zone
- Each general purpose input can be programmed as TTL, high range or monitored high range
- General purpose inputs allow monitoring for short to ground and open circuit
- Up to 500 virtual inputs via RS232 or Ethernet
- Provides system health monitoring via RS232 or Ethernet
- Status LEDs
- Rack mountable (1RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked and RoHS compliant
- Covered by Biamp Systems' five-year warranty



The CI-1 Control Interface is a companion product to the Vicia LSI-16 Life Safety Interface. It facilitates necessary connections to the LSI-16/LSI-16e to meet EN 54-16 standards.

- High reliability switches for Local Sounder Silence, System Test and System Fault Reset
- High level sounder for Fault/Alarm warning
- Dual power summing with power loss fault connection
- Provides terminating resistors for Alarm and Fault Inputs
- Provides terminating resistors for any unused monitored outputs
- Current limited reference voltage output
- Rack mountable (1RU)
- EN 54-16 certified, EN 60849 & AS 60849 verified, CE marked and RoHS compliant
- Covered by Biamp Systems' five-year warranty

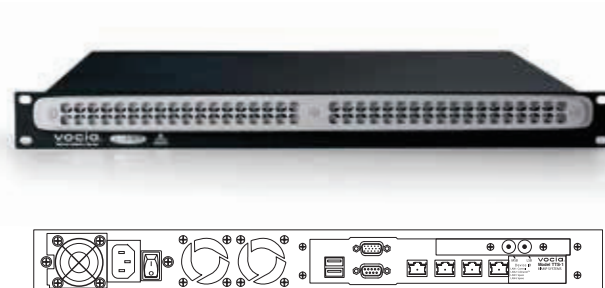
CI-1
VOCIA CONTROL INTERFACE
EN 54-16 CERTIFIED



The MS-1 is a networked message server that supports multiple paging functions within a Vicia system—including message playback, event scheduling, VoIP paging interface, logging and remote access.

- Recorded message storage and playback
- Event scheduling
- Storage of logged system data
- VoIP paging interface
- Inter-world paging support
- System configuration storage and service
- Time server support
- CobraNet audio/control with dynamic use of available bundles over single Ethernet cable
- Remote third party control capability
- Status LED
- Separate Ethernet ports for TCP/IP Control, CobraNet and VoIP
- Rotary switches for unit identification
- Rack mountable (1RU)
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' three-year warranty

MS-1
VOCIA MESSAGE SERVER

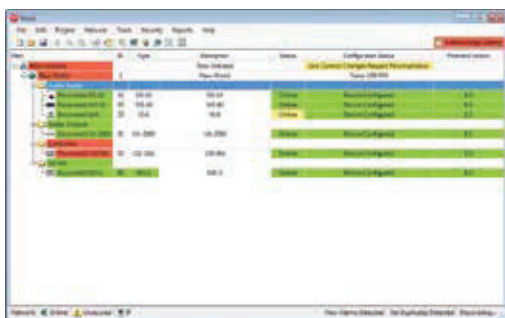


The TTS-1 and the TTS-1nc are designed to work in conjunction with a Vicia Message Server 1 (MS-1) to enable text to speech messaging as part of a Vicia system solution. The TTS-1 and TTS-1nc use Ethernet-based control protocols in conjunction with CobraNet to function within a Biamp Vicia system and constructs announcements using a set of user-defined templates.

- Text-to-speech announcement creation from any computer with appropriate network access
- Announcements in multiple languages and voices
- User-defined templates
- CobraNet audio/control with dynamic use of available bundles over single Ethernet cable
- Status LED
- Rotary switches for unit identification
- Rack mountable (1RU)
- CE marked, UL listed and RoHS compliant
- Covered by Biamp Systems' three-year warranty

TTS-1 and TTS-1nc
VOCIA TEXT-TO-SPEECH SERVER

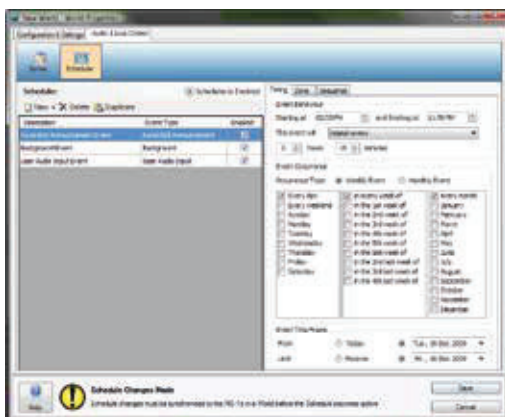
SOFTWARE



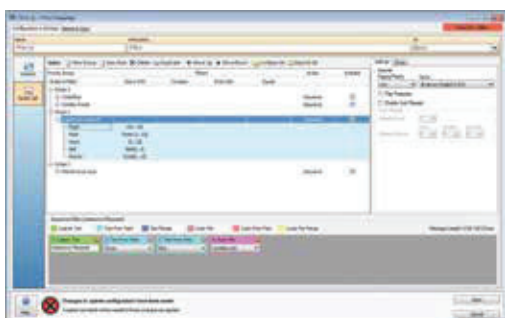
The main screen of the Vocia software provides an easy, at-a-glance overview of your universe, whether it contains a single world or several. This hierarchical view lists all devices included in the Universe by World, Device ID, online and configuration status, along with the loaded firmware version. The colors indicate the health of each device listed. An overall system alarm tab ("Acknowledge Alarms" in the upper right corner) flashes when an alarm or fault is detected in the system.



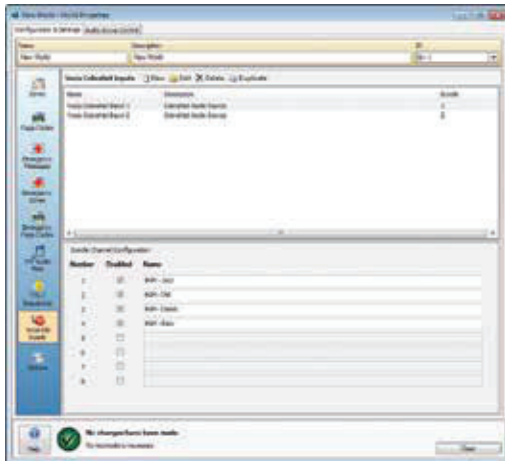
The software is designed for ease-of-use and intuitive interfacing. Most of Vocia's devices offer fixed-chain DSP parameter adjustment and live control. This screen shows the properties sheet of the VI-6 with its audio and live control settings; filters are applied to the input levels, representing one possible configuration. The filters are displayed as individual lines in the EQ filter graphic.



The scheduler feature (requires MS-1 to be available) (found under world properties by double clicking on the world icon) is used to configure schedules for recorded announcements (MS-1 required), background events (e.g. background music coordinated with the time of day) and user audio input events (e.g. an announcement that broadcasts for a certain period of time in a specified zone).

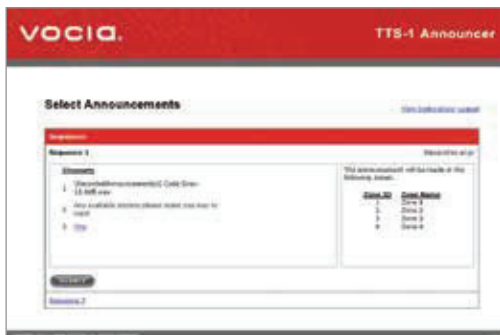


The TTS-1nc Text-to-Speech Nursecall Middleware server supports the same Windows[®] client that is used to enter text segments for speech conversion as the standard TTS-1. However, the TTS-1nc server supports a direct RS-232 serial interface to Rauland-Borg Responder 4 or 5 pocket page servers to allow automatic text conversion. It also provides a software user interface that allows graphical configuration of the Middleware and Matching Rules Engine that is available on the TTS-1nc.



Features of the Vocia software are designed to interface to any CobraNet audio system using explicit bundle numbers. Now you can expand existing audio systems and add Vocia functionality for critical paging and voice evacuation.

NOTE: A VO-4 or VO-4e is required for each four CobraNet outputs from Vocia into Audia. For example, if 10 outputs are required from Vocia into Audia, three VO-4s or VO-4e's will be required to establish the outputs. No additional hardware is required for inputs to Vocia from Audia.



The Vocia TTS-1 Text-to-Speech server supports a windows client that is used to enter the text segments for speech conversion. The LSI-16/ LSI-16e supports a browser interface for remote monitoring of emergency faults.

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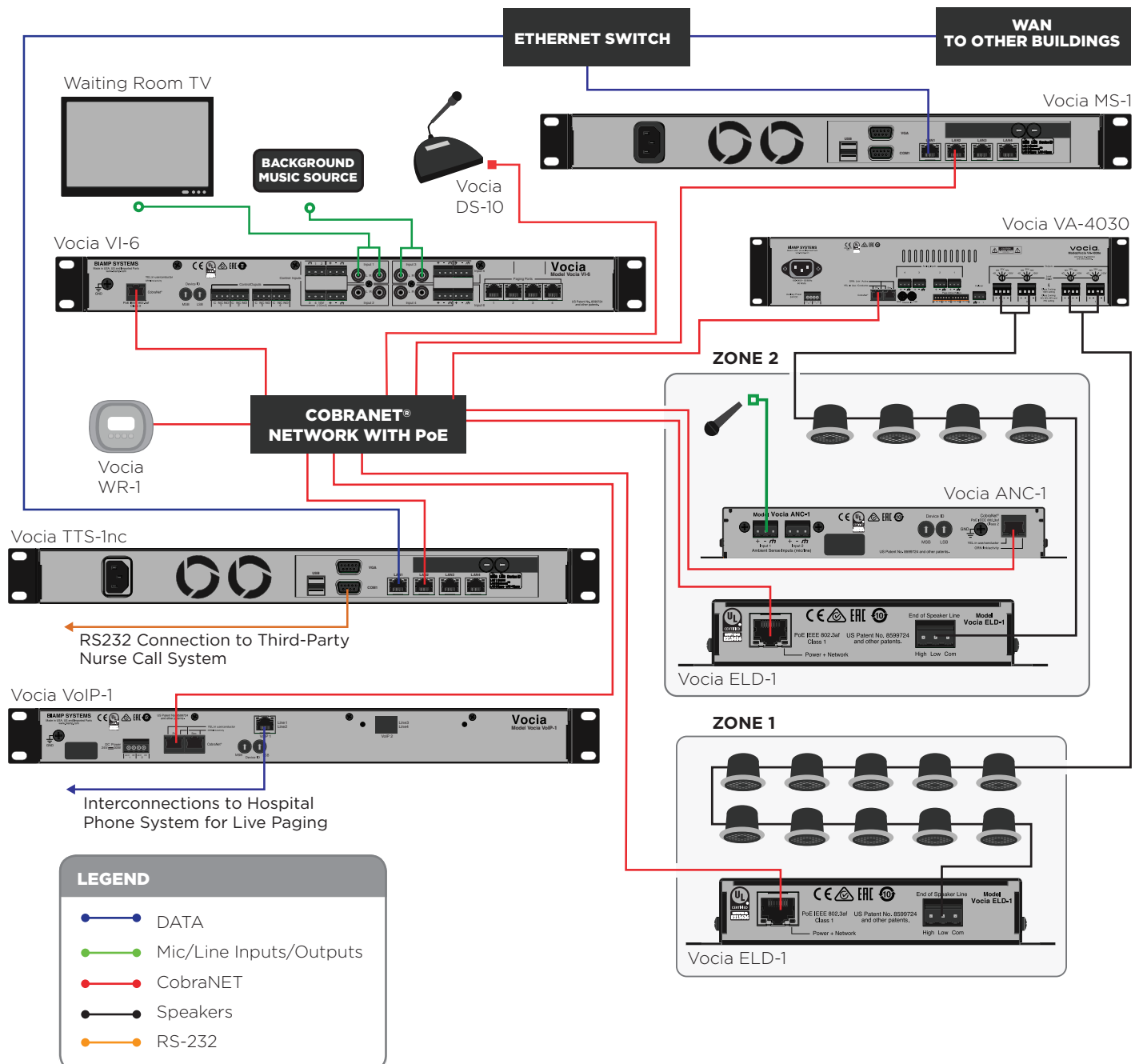


SYSTEM DESIGN GUIDE HEALTHCARE

VOCIA® NURSE CALL INTEGRATION

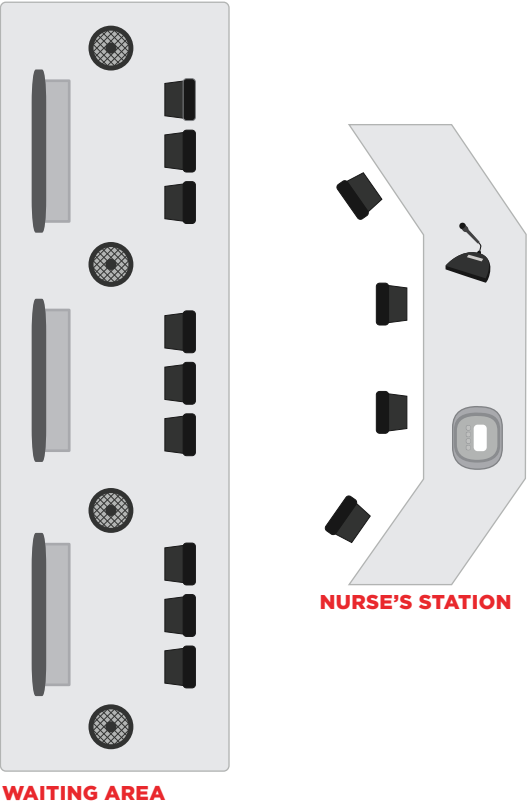
The unique need of healthcare facilities for professional-grade networked public address, and voice evacuation systems requires a very specialized audio solution. With Vicia, the decentralized architecture distributes audio processing and page-routing across the network, eliminating the potential for a single point of system failure. The Nurse Call integration package enables a direct connection to supported Nurse Call systems such as Rauland-Borg.

HERE'S HOW VOCIA WORKS ACROSS A SMALL HEALTHCARE FACILITY:

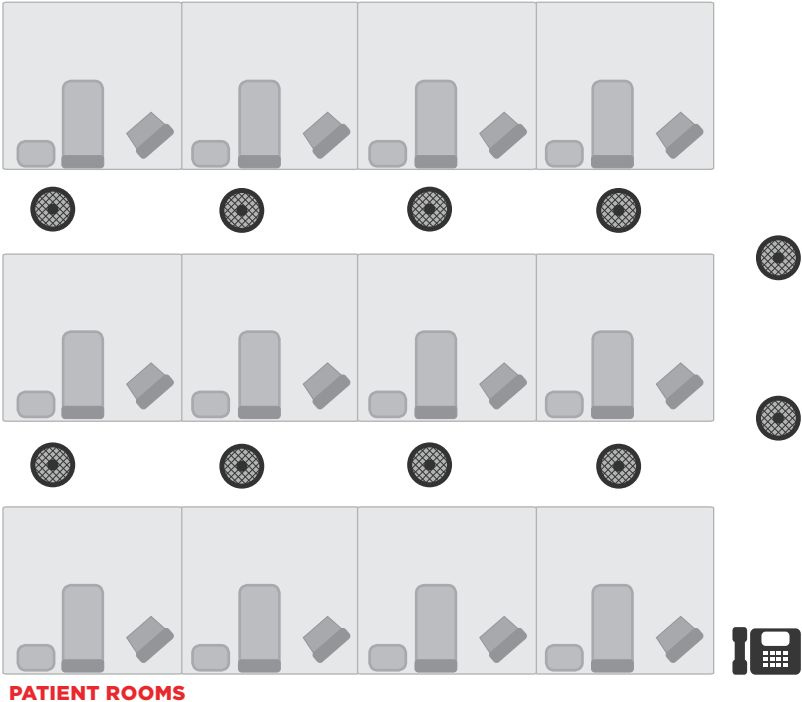


HEALTHCARE FACILITY FLOOR PLAN

ZONE 2



ZONE 1



KEY

 Ceiling Speaker

 DS-10

 VR-1

 VoIP Telephone with Paging

EQUIPMENT LIST

Quantity	Product	Function
1	DS-10 Desk Station	Nurse's paging station with embedded DSP and up to 999 user-configurable page codes.
1	VI-6 Vocia Input Device	Allows up to 6 channels for background music via CobraNet.
1	VA-4030 Amplifier	Provides amplification to all speakers.
1	ANC-1 Ambient Noise Compensation Device	Adjusts the volume of the pages depending on the ambient noise in the zones.
2	ELD-1 End of Line Device	Monitors each zone of speakers and automatically reports shorts or opens in the speaker line.
14	Ceiling-mounted speakers	Provides sound reinforcement for patient room zones and waiting area zone.
1	MS-1 Message Server	Allows multiple functions including message playback, event scheduling, and paging to other Vocia worlds.
1	TTS-1nc Text-to-Speech: Nurse Call Device	Provides ability to interact with a third-party Nurse Call solution such as Rauland-Borg.
1	VR-1	Controls volume of background music.
1	VoIP-1	Allows real-time VoIP paging within the Vocia system.



SYSTEM DESIGN GUIDE

MEDICAL CENTER

Vocia®

Product	Function
VA-8150 CV Amplifier	Provides network amplification for the first floor of the Vocia system.
VA-4300 CV Amplifier	Provides network amplification for the 2nd and 3rd floors of the Vocia system.
ANC-1 Ambient Noise Compensation devices	Controls output levels making automatic adjustments in response to changes in ambient noise levels.
DS-10 Paging Stations	Desktop networked paging stations with up to 999 user-configurable page codes.
EWS-4 Emergency Paging Station	Wall-mounted networked paging station with four software configurable emergency paging levels and four user-configurable page codes.
VI-6 Audio Input Device	Allows up to six local audio sources, such as background music, to be added to the Vocia system.
VI-8 Input Device	Allows emergency messages to play directly from the Fire Alarm Control Panel through the Vocia system.
MS-1e Message Server	Offers multiple paging options, including a VoIP paging interface, message playback, and event scheduling.
LSH-16 Life Safety Interface	Serves as an interface between a Vocia system and the Fire Alarm Control Panel.
CI-1 Control Interface	Allows the LSH-16 to interface with the Fire Alarm Control Panel.
TTS-Ince Text-to-Speech Server	Creates text-to-speech announcements and integrates with nurse call solutions.

VOCIDG-449-1605-EN-RT

As technology advances and access to healthcare grows, medical centers worldwide are facing more demand than ever. Patients deserve the highest quality care in order to recover and thrive. Most modern medical facilities serve several different functions, from Birthing Centers to the Emergency Department to the Intensive Care Unit. Because such a diverse array of patients and medical professionals are onsite each day, healthcare providers are particularly sensitive to the importance of a flexible and comfortable healing environment. That means ensuring that pages intended for a physician in a particular department don't disturb patients who are recovering in a different area of the building, and that important announcements reach the appropriate floors. In this Healthcare scenario, different medical departments require specific paging and communication capabilities to meet the needs of staff, patients, and visitors.

SYSTEM DESIGN GUIDE

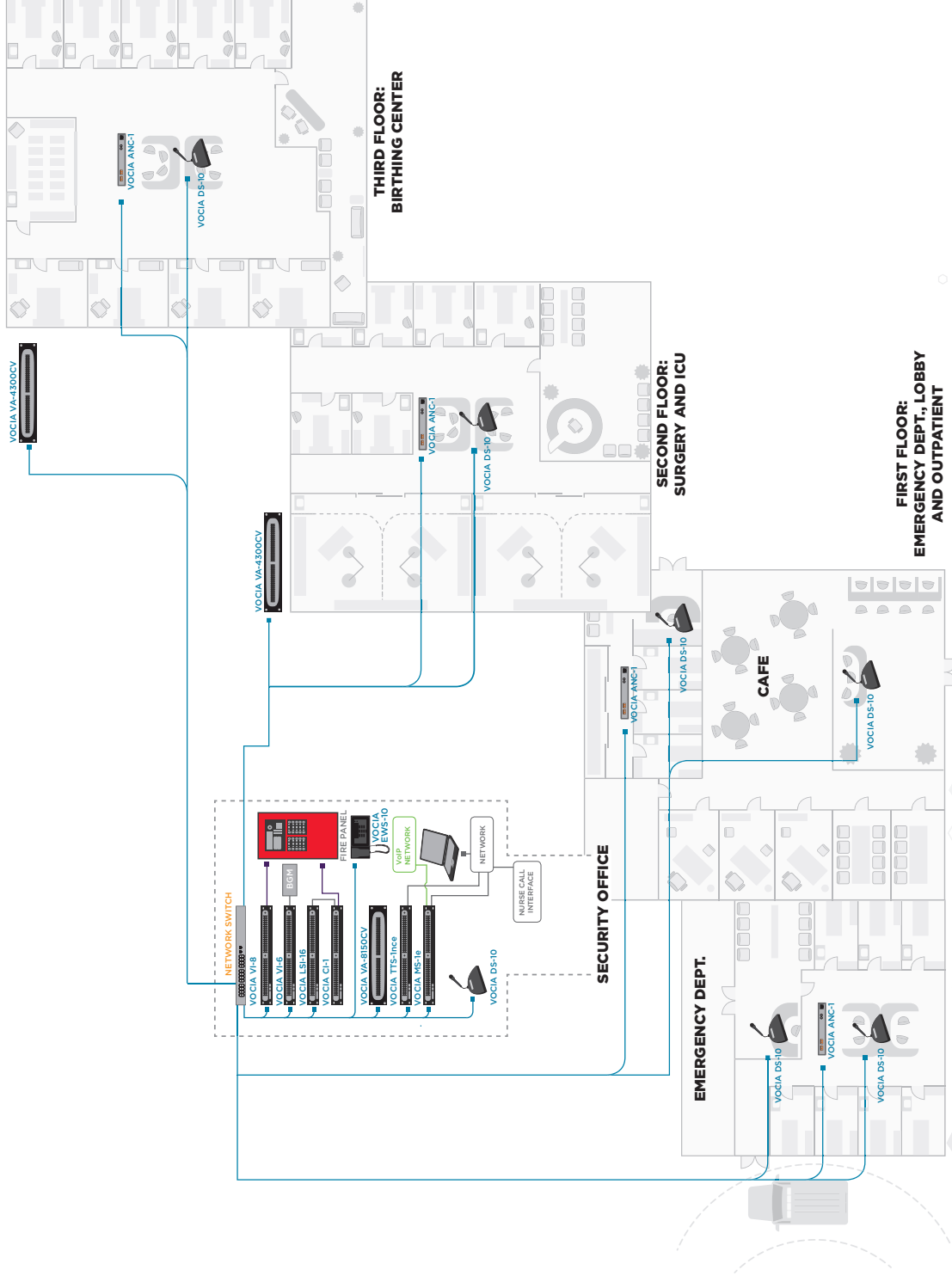
MEDICAL CENTER

Biamp's Vocia platform can perform a number of standard communication functions for day-to-day hospital activities in each department, from nurse call announcements to paging to emergency communications. Because different floors and departments have particular needs, Vocia supports zoned paging to communicate exactly where the messages are needed without disrupting other areas of the medical center.

A quieter healing environment contributes to faster recovery for patients. That's where Vocia's Ambient Noise Compensation (ANC) technology comes in. Integrating paging with ambient noise compensation allows the paging volume to adjust automatically to the space's ambient volume, ensuring pages are audible and intelligible, without unnecessary disruption of the healing process. In addition, Vocia's networked paging can interface with nurse call software, which also contributes to a quieter healing environment. Furthermore, medical centers must continue providing care in the event of a power outage or the loss of part of the network. Vocia's decentralized network means there's no single point of potential failure, allowing the system to continue functioning normally if one portion goes offline.

VOCIA FEATURES

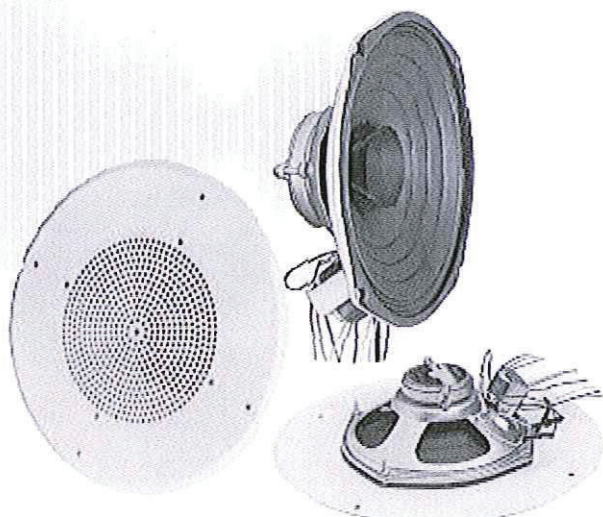
- Decentralized networking with no single point of failure
- Standard paging, zoned paging, and critical paging in one platform
- Seamless integration with nurse call solutions, as well as audio conferencing solutions like Tesira
- Scalable to grow with a facility's needs



Speakers - Horns - Speaker Accessories



MODEL: ACC1403 - 8" SPEAKER ASSEMBLY W/VOLUME CONTROL



Simple
Volume
Adjustment

SPECIFICATIONS

BAFFLE

Similar to Rauland ACC1000:

See Catalog Sheet for detailed specifications

SPEAKER

Type: 8" 920.32 cm) PM

Frequency Range: 65 to 17,000Hz

Power Rating: 8 watts RMS

Sensitivity: 93dB at 4' with 1 watt input

Magnet Weight: 5.0 oz. (141.75 g) ceramic

FEATURES

- Preassembly Save Installation Time, Money
- With 25V/70V Line Matching Transformer
- Includes Built-In Volume Control
- "Whizzer" Cone for Extended High Frequency
- Features Full 5-Ounce Ceramic Magnet

Voice Coil Impedance: 8 ohms

Voice Coil Diameter: 0.75" (1.91 cm)

Flux Density: 9,500 gauss

Transformer: 25V/70V line; taps 5/16, 5/8, 1-1/4, 2-1/2, 5 watts

Volume Control: 50-ohm potentiometer

Depth: 2.75" (6.99 cm)

Finish: Baked enamel

Weight Complete Assembly: 2 lbs. 11 oz. (1.22 kg)

DESCRIPTION

The Rauland ACC1403 Speaker Assembly consists of a Rauland 8" 5-ounce magnet speaker complete with 25V/70V line matching transformer, assembled to a Rauland ACC1000-type baffle. Includes a built-in volume control easily adjustable by screwdriver through the center aperture of the baffle. Each component in the assembly

is a high quality unit. The loudspeaker with its added "whizzer" cone for extended high frequency response, delivers exceptional tone quality with excellent output. The one-piece steel speaker baffle in an attractive semigloss white finish harmonizes with any interior.

ASSOCIATED EQUIPMENT

The Speaker Assembly shall be a Rauland ACC1403 or approved equal. It shall consist of a Rauland 8" 5-ounce magnet speaker complete with 25V/70V line matching transformer and built-in 50-ohm volume control, assembled to a Rauland ACC1000-type Baffle. The circular baffle shall be 12-7/8" (32.70 cm) in diameter. It shall be constructed of 22-gauge cold-rolled steel (Zincrometal), with a baked-on semi-gloss white epoxy finish which shall be virtually mar-proof. The assembly shall be screw-mounted in any standard backbox.

The speaker shall be an 8" cone-type with the following specifications: Frequency response shall be 65 to 17,000Hz, ceramic

magnet weight shall be at least 5 ounces (141.75 g); axial sensitivity shall be 93dB at 4' with 1 watt input; normal wattage rating shall be at 8 watts RMS; "whizzer" cone for extended high frequency shall be included; voice coil impedance shall be 8 ohms and shall measure 0.75" (1.91 cm) in diameter; speaker depth shall be 2.75" (6.99 cm). The speaker shall be equipped with a universal line matching transformer for a 25V/70V output line with taps at 5/16, 5/8, 1-1/4, 2-1/3 and 5 watts. The volume control shall be a 50 ohm potentiometer which shall be screwdriver adjustable through the center aperture of the baffle.

The entire assembly shall weight 2 lbs. 11 oz. (1.22 kg).



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Architect and Engineer (A&E) Specifications
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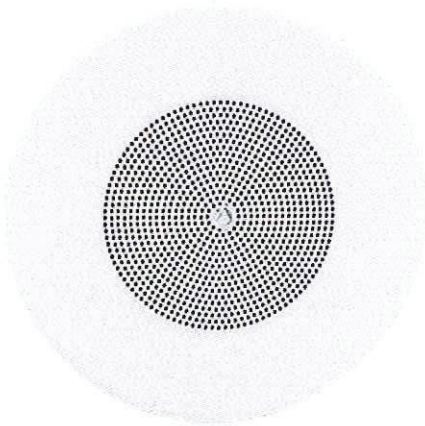
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Choices include mounting style, magnet size, transformer & grille type for installations that do not require back boxes.

8" Dual Cone In-Ceiling
Loudspeaker with 25V/70V
5-Watt Transformer and 62-8
Baffle

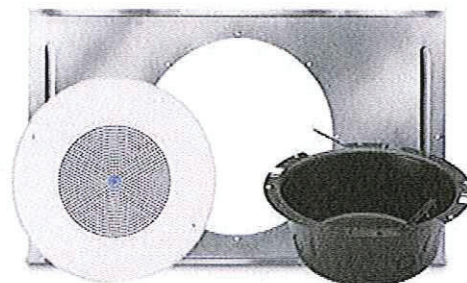


SD72W

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Qty 4 SD72 Loudspeaker, 62-8
Baffle, CS95-8 Back Box, and
81-8R T-Bar Bridge Kit



SD72W-KIT4

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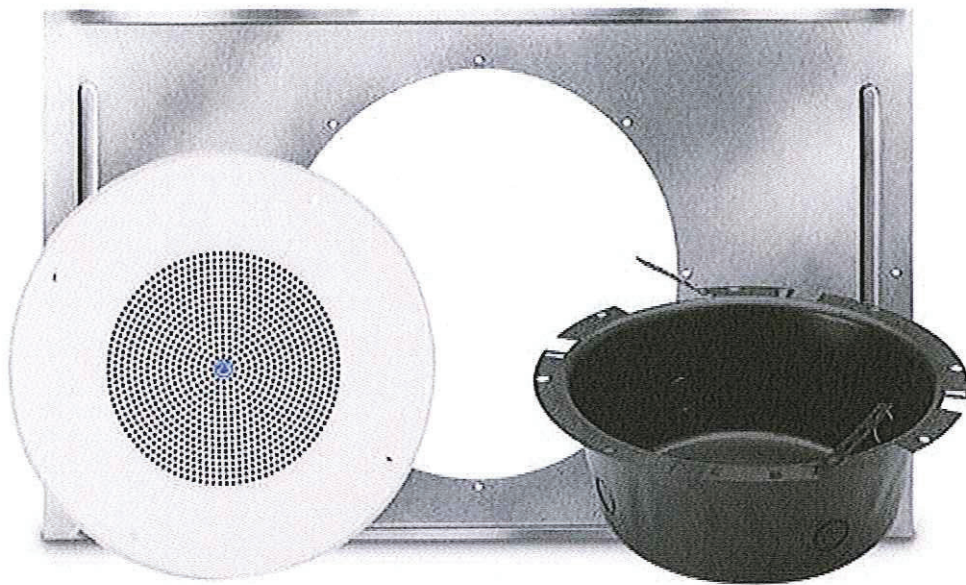
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[SD72W-KIT](#)



SD72 LOUDSPEAKER, 62-8 BAFFLE, CS95-8 BACK BOX, AND 81-8R T-BAR BRIDGE KIT

[SD72W-KIT](#)

AtlasIED offers the SD72W-KIT to make ordering a complete solution easy for in-ceiling loudspeaker applications. The kit includes products that have been selected to work perfectly together in BGM and paging applications.

SD72W-KIT includes:

- [SD72W](#) 8" Dual Cone In-Celing Loudspeaker with 25V/70V 5-Watt Transformer and 62-8 Baffle
- [CS95-8](#) 8" Recessed Round Stacking Enclosure
- [81-8R](#) 8" T-Bar Tile Bridge

The loudspeaker used in the SD72W is the next generation of Atlas high performance cone loudspeakers, very similar to the preceding C-Series loudspeakers. The similarity in the sonic qualities ensures that contractors and designers who have worked with the AtlasIED C-Series in the past will know exactly what to expect when they use the SD72W.

Model SD72W is a dual cone, 25 watt 8" (205mm) loudspeaker with a 10oz. (260g) ceramic magnet. It includes a curvilinear, treated paper cone for lower harmonic distortion. The loudspeaker is also equipped with a full 1" diameter copper voice coil with a black anodized aluminum former for better power dissipation. Package includes factory installed 25/70.7V line matching transformer with tap selections ranging from .25 to 5 watts.

Model SD72W operates within a frequency response range of 45Hz - 19kHz (nominal) with a sensitivity of 97dB and a dispersion angle of 105°.

The SD72W includes a factory installed CRS baffle with concealed loudspeaker mounting studs and includes screw mount hardware. The SD72W baffle features standard mounting holes to accommodate the majority of AtlasIED 8" tile bridges, mounting rings and enclosures.

The CS95-8 enclosure is furnished with four combination conduit knockouts 1/2" – 3/4" (13-19mm), four J-clips and two flexible perforated mounting straps. One-piece CRS construction, undercoated, patch-jute lined, and finished in epoxy.

The 81-8R tile bridge features a round cut-out designed to work with the CS95-8 enclosure. The unit is constructed of 24-gauge CRS with an electrogalvanized rust-resistant finish.

SD72W-KIT MUST BE ORDERED IN MULTIPLES OF 12.

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TECH SPECS

SPEAKER

SPEAKER TYPE Treated Paper

SENSITIVITY 97dB Average

FREQUENCY RESPONSE 55Hz - 8kHz

POWER TAPS .25, .5, 1, 2, & 5 Watts

DISPERSION 105° (2kHz Octave Band, -6dB Points)

DIAMETER 8 1/8" (20.6375 cm)

SAFETY LISTED 1480, 2043

SUPPORT FILES

CAD DRAWING SD72W

DOWN

INSTALLATION SHEET ATS004381 SD72, SD72W, SD72WV Install Sheet (1 page)

VIEW | DOWN

REVIT DRAWING SD72W Revit Drawing

DOWN

<  >

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Quam

LOUDSPEAKER ASSEMBLIES

C10X/BU/WS

CEILING MOUNTED

8" O.D. Loudspeaker with a 10 oz. Magnet,
5W-25/70V Transformer, Stud-Mount Baffle

GENERAL DESCRIPTION

The **C10X/BU/WS** consists of an 8C10PAX - 8" O.D. dual cone loudspeaker with a 10 oz. magnet, TBLU - 5W-25/70V transformer, and a BR8WS - round, stud-mount baffle in a white powder coat finish.

INTENDED USE - Indoor Environment

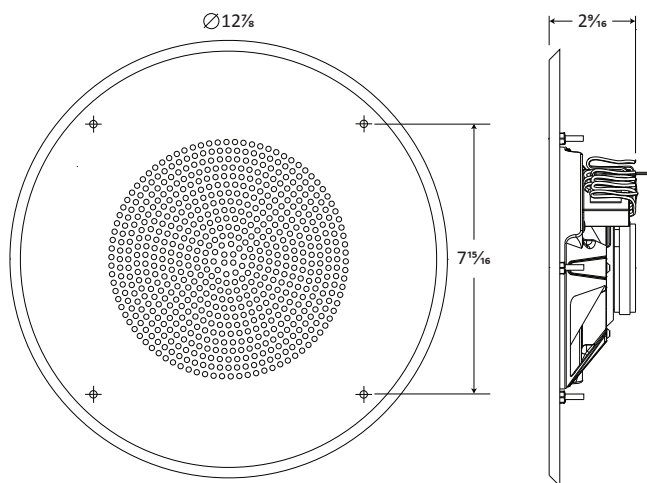
Program Material	Signal tones, voice, and music
------------------	--------------------------------

AUDIO SPECIFICATIONS

Average Sensitivity	95 dB SPL, 1W/ 1M
Loudspeaker Power Rating	20W RMS, EIA 426A Standard
Maximum Power Rating	15W @ 8 Ohms
Calculated Output	102 dB SPL, 5W/ 1M
Magnet Type & Weight	BeFe Ceramic, 5 oz. Nominal
Frequency Response	65 Hz - 17 kHz EIA 426A Standard
Nominal Coverage Angle	100° Included Angle, -6 dB / 2 kHz, Half space
Audio Connection	7" Color-Coded Leads
Wattage Taps	0.32W, 0.63W, 1.25W, 2.50W, 5W

RECOMMENDED QUAM COMPONENTS

Enclosures	ERD-8, ERD8U
Mounting Support	SSB-2, SSB-3, SSB-7



Recommended construction sequences, installation instructions, and A&E Specifications are available in various formats in the Specs and Diagrams section at: www.QuamSpeakers.com

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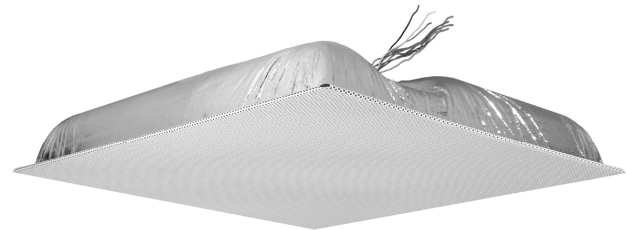
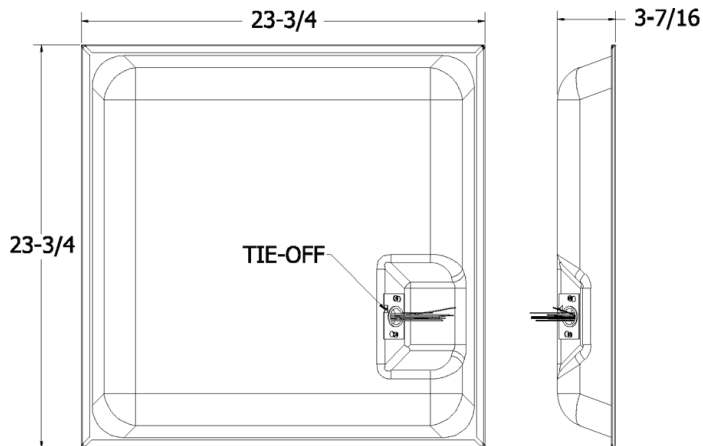


Quam

SYSTEM 15

Tile replacement Loudspeaker Assembly

0104.5 - LAY-IN CEILING MOUNT (TILE REPLACEMENT)



GENERAL DESCRIPTION

The Quam SYSTEM 15 is a complete loudspeaker assembly, with 8" frame, wide frequency response, coaxial loudspeaker, 70.7V transformer, perforated steel baffle and integral back box.

Finish options: White or black baked epoxy hybrid.

INTENDED USE

Mounted parallel to floor plane in a 24" wide suspended ceiling tile grid; indoor environment.

Program Material: Music, voice and signal tones

SYSTEM COMPONENTS:

Loudspeaker	8C10FECO
Transformer	TBH70, 5 taps at -3dB levels
Finish options	Order as SYSTEM 15/B
Black only	

AUDIO SPECIFICATIONS

Speaker Size:	8" frame
Average Sensitivity:	97 dB-SPL, 1W/ 1M
Loudspeaker Power Rating:	20W-RMS, EIA 426A Standard
Calculated Output:	103 dB-SPL, 4W / 1M
Frequency Response:	40 - 20kHz EIA 426A Standard
Nominal Coverage Angle:	100° Included angle, -6dB 2 kHz, half Space
Audio Connection:	7" long, color coded tinned wires, pre-cut

Recommended construction sequences, installation instructions and A&E Specifications are available in various formats on the Quam Architects' and Engineers' Resources CD or on the web at: www.quamspeakers.com

RECOMMENDED ADDITIONAL QUAM COMPONENTS

N/A

Quam-Nichols Company

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01

LOUDSPEAKER ASSEMBLIES

H20-G

Compact compression driver horn



The H20-G is a powerful weatherproof industrial horn loudspeaker with compression driver. It offers a high sound pressure, which makes this loudspeaker ideally suited for speech and the reproduction of alarm signals in a noisy environment. The H20-G is ideal for long-distance and outdoor use.

The stainless steel brackets and rust free bolts and screws make this horn suited for indoor and outdoor use. The bracket also makes it possible to rotate the speaker over a range of more than 180 degrees.

Because this horn speaker is constructed in high impact ABS, it is able to withstand even the harshest of elements.

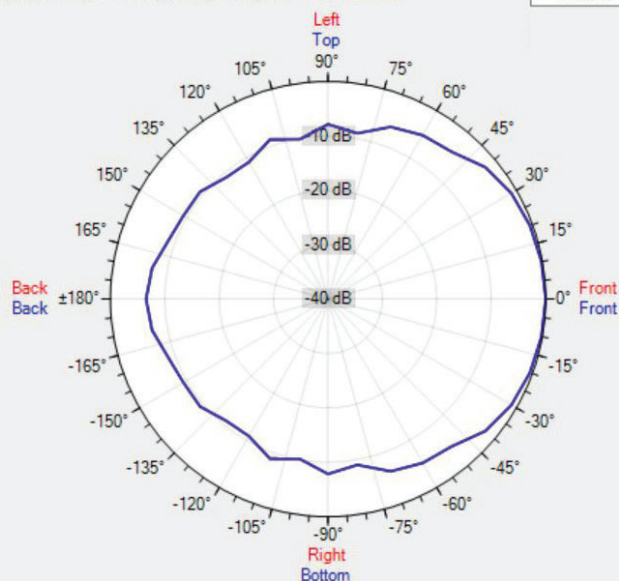
The H20-G loudspeaker is equipped with a high quality 100 volt transformer with several power taps, and can also be used in 8 ohms systems. Our technology avoids any overload of your amplifier by controlling the impedance with the unique IMC protection device inside.

TECHNICAL SPECIFICATIONS

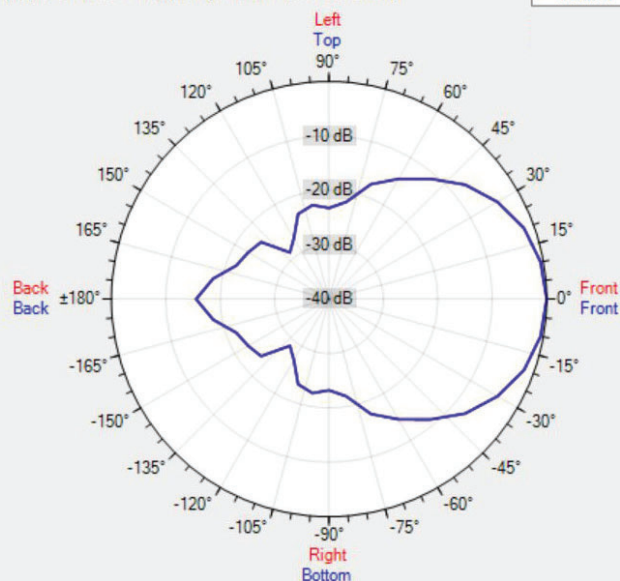
outside diam. in mm	200	depth in mm	230
loudspeaker system	compression driver	mounting system	U-bracket
colour	grey	100V transformer power taps in watts	20 - 10 - 5 - 2.5
impedance in ohms	8	low impedance dynamic power in watts	30
low impedance RMS power in watts	20	SPL 1W/1m in dB	105
max SPL 1m in dB	116	frequency response in Hz	280 - 9.5 k
main construction material	ABS plastic	IP rating	66
applicable low impedance	yes	applicable in 100V	yes
Vertical dispersion angle 1000 Hz	180°	closest RAL colour (subject to deviations)	RAL9006
Net weight product (kg)	1.36		

POLAR PLOTS

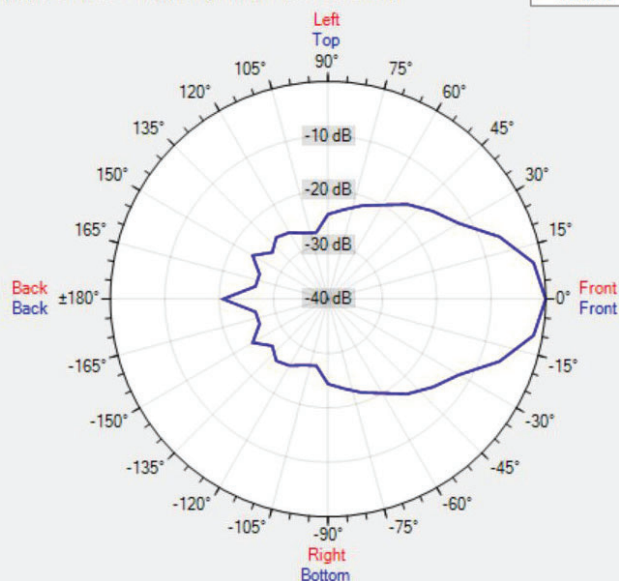
Data Shown: H20 (Apart Audio)
Display Parameters: Frequency: 1000Hz (1/24th Octave)



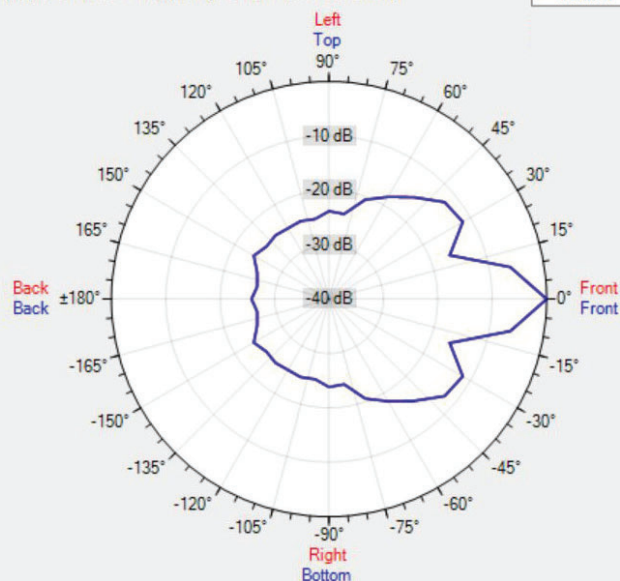
Data Shown: H20 (Apart Audio)
Display Parameters: Frequency: 2000Hz (1/24th Octave)



Data Shown: H20 (Apart Audio)
Display Parameters: Frequency: 4000Hz (1/24th Octave)



Data Shown: H20 (Apart Audio)
Display Parameters: Frequency: 8000Hz (1/24th Octave)



[MORE PICTURES](#)



Appendix E

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Submittal Register 60% DD Submission

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